

UFOR 511: Geomatics Principles and Applications

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2026-09-16

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Welcome

These are the course materials for UFOR 511 in the Master of Urban Forestry Leadership (MUFL) at the University of British Columbia (UBC). These Open Educational Resources (OER) were developed to foster the Geomatics Community of Practice that is hosted by the Faculty of Forestry at UBC.

These materials are primarily lab assignments that students enrolled in UFOR 511 will complete and submit for credit in the program. Note that much of the data referenced are either public datasets or otherwise only available to students enrolled in the course for credit. Deliverables for these assignments are submitted through the UBC course management system and only students enrolled in the course may submit these assignments for credit.

How to use these resources

Each “chapter” is a standalone lab assignment designed to be completed over one or two weeks. These assignments are designed for emerging urban forestry leadership professionals who seek to expand their skillset in fundamental geomatics concepts as applied to the field of urban forestry.

Students enrolled in UFOR 511 will submit all deliverables through the course management system at UBC for credit and should consult the schedule and deadlines posted there. The casual user can still complete the tutorials step-by-step, but the data that are not already publicly available are not hosted on this website and therefore you will not have access to them.

Unless otherwise noted, all materials are Open Educational Resources (OER) and licensed under a Creative Commons license (CC-BY-SA-4.0). Feel free to share and adapt, just be sure to share with the same license and give credit to the author.

How to get involved

Because this is an open project, we highly encourage contributions from the community. The content is hosted on our GitHub repository and from there

you can open an issue or start a discussion. Feel free to open an issue for any typos, factual discrepancies, bugs, or topics you want to see. We are always looking for great Canadian case studies to share! You can also fork our GitHub repository to explore the source code and take the content offline.

Chapter 1

Exploring Spatial Data

Written by Paul D. Pickell

Lab Overview

In this lab you will learn how to find, evaluate, and visualize vector spatial data. You are encouraged to select datasets for a geography that is meaningful to you, such as the area where you are planning to do your project or capstone for your program. This will help you begin to build a library of data that may be directly useful later. As you work through the lab, you will explore some of the basic capabilities of the ArcGIS Pro software that we will continue to build upon throughout the course. You will look at a variety of different spatial data file formats and data types.

Learning Objectives

- Locate and download relevant datasets from open data portals
- Identify spatial data file formats and attribute field data types
- Inspect metadata (coordinate reference system, extent, file type, data type) within ArcGIS Pro
- Choose the most appropriate coordinate reference system for your study area
- Examine attribute tables to understand data content, codes, and field data types
- Create a simple map that visualizes multiple vector spatial datasets

Deliverables

Responses to the questions posed throughout the lab on the course management system. (75 points)

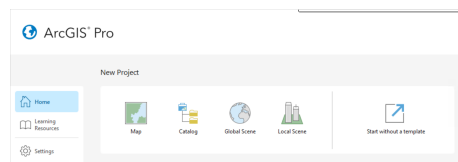
Upload your final map that shows your three selected datasets. At least one dataset must be symbolized by a field in the attribute table. You can pick the colors/symbols to use for each layer. Layers that appear to use the default random color/weight assignment by ArcGIS Pro will not be accepted for credit. You must show that you manipulated the symbology for each layer. Your map must also contain a title, north arrow, scale bar, and a legend that contains the symbologies of all the layers. (25 points)

Data

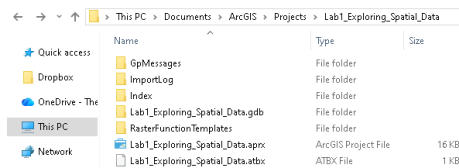
For this lab, you will explore different publicly available spatial data portals.

Task 1: Create an ArcGIS Pro Project

Step 1: Start ArcGIS Pro. On the start-up screen, you will be prompted to select the type of project. Select “Map” and give your project a meaningful name, something like `Lab1_Exploring_Spatial_Data` is a good choice. When you create a project, ArcGIS Pro will create a folder with your project name in `C:\Users\[your username]\Documents\ArcGIS\Projects\[your project name]\`.

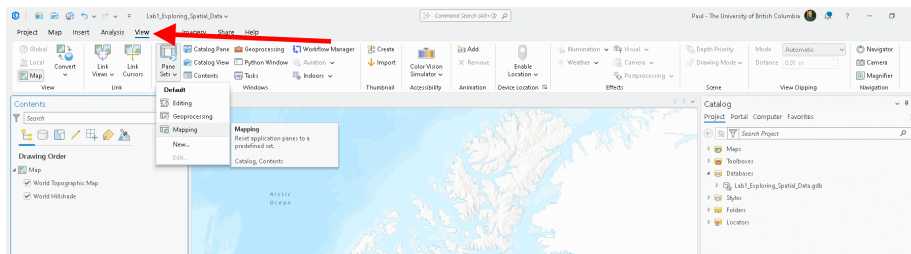


Inside this folder, you will find ArcGIS Pro has generated a bunch of files and folders. Importantly, there is the `.aprx` file, which is the file you can use to open your project in ArcGIS Pro. The other important folder is `FRST538_Lab1.gdb`, which is the default geodatabase for your project. When you start running different tools in ArcGIS Pro, this is where the outputs will automatically be stored. If you navigate to this folder in Windows Explorer, it will just look like a bunch of random files, so you can only view the contents of your geodatabase through GIS software like ArcGIS Pro.



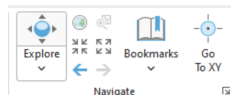
Take note of your project location because this is where you will be saving the spatial data files that you download so that you can view them in ArcGIS Pro.

In ArcGIS Pro, you should take note of four areas: - The map in the middle; this is where your spatial layers will be mapped, right now you are looking at the default basemap. - The navigation ribbon at the top (Project, Map, Insert, Analysis, View, Edit, Imagery, Share, Help); this is where you can add data and find tools. - The “Contents” pane on the left; this is like a table of contents where your spatial data layers will be listed in the “top-down” that they are drawn/appear on the map. - The “Catalog” pane on the right; this lists sources for data and if you expand “Databases”, you will see your default geodatabase listed there that has the same name as your project.



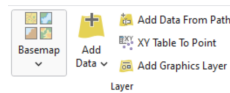
If you do not see the “Contents” and/or “Catalog” panes, then navigate to the “View” tab from the top ribbon, “Set Panes”, then select “Mapping” to return to the default configuration.

Step 2: Navigate to the “Map” tab from the top ribbon. This will be the most useful tab for this lab as it allows you to navigate around the map, add data, and select/query the spatial data.



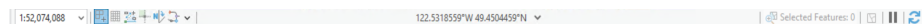
The “Navigation” group has methods for panning around and zooming in/out of the the map. If you click “Explore” then you can pan freely around the map. Using your mouse scroll wheel or trackpad scroll gesture on your computer, you can zoom in/out. the other buttons with the arrows allow you to zoom in/out of the centre point of your current map view at fixed intervals. “Bookmarks” allow you to return to the current view of the map after you have navigated away. These can be really handy for returning to the extent/scale that you previously viewed something or maybe for re-locating a feature that you found interesting.

If you have a specific coordinate, you can use “Go To XY” to automatically pan to that location.



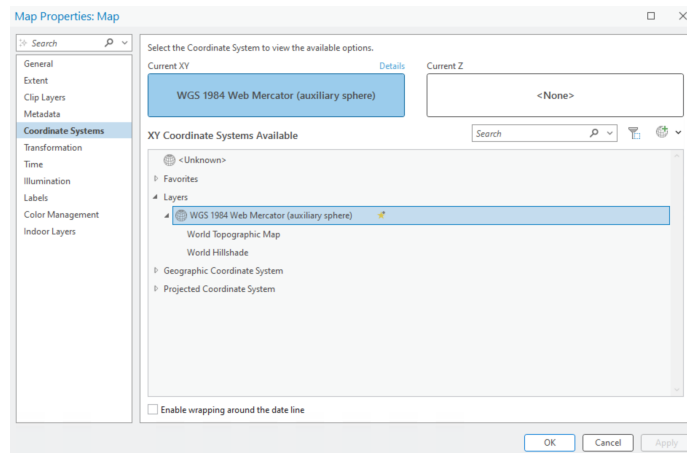
The “Layer” group allows you to change the basemap and add spatial data as layers from various sources. An important concept to understand in GIS is the distinction between “layer” and “data”. “Layers” are exactly that, they are what we *see* in the map. We can change the colours and symbology of layers in ArcGIS Pro, and these *visual* choices are stored as part of the `.aprx` project file. When you add spatial data to your map, they *appear* as “layers” in the “Contents Pane”. On the other hand, “data” are the spatial data files that are being read from your project folder or some other source. When you change the symbology of a layer, we are not editing or changing the data in any way. Even if remove the layer from your map, you are not deleting the data file, rather you are removing the *reference* to that file from your project that is used to display the layer.

Q1. What are the names of the current layers in your map? (4 points)



We will explore the other tool groups in later tasks and other labs. For now, the last feature that you should be aware of is the information toolbar at the very bottom of the application. Here you will find the current scale of your map (1:52,074,088 in the image above), and you can also type any number directly into the box and press Enter to change to a different, set scale. In the middle of the toolbar you can see the location of your cursor with the default units in decimal degrees. On the far right, there is a pause button and a refresh button. These are used to stop loading or re-load the layers in your map. These can be very useful later when it might take your computer a long time to re-draw thousands of features in the layers in your map. In this case, you can pause drawing mode momentarily while you configure the layer symbology.

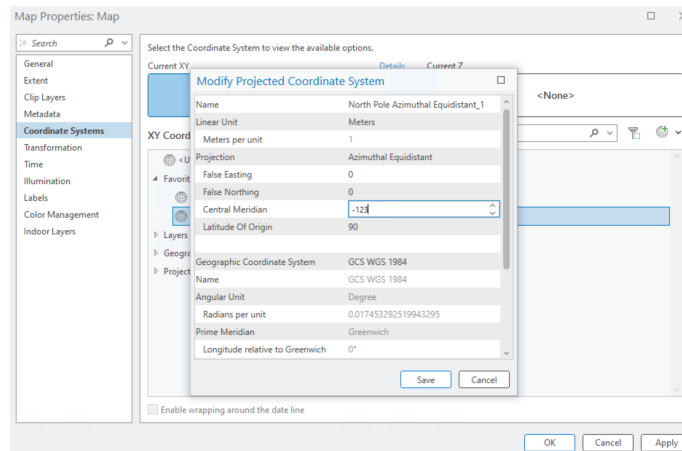
Step 3: From the “Contents” pane, right-click on “Map”, and select “Properties”. Here you can find and set various properties about your map. Navigate to the “Coordinate Systems” from the list on the left and you will see the current Coordinate Reference System (CRS) that is being used by your map.



Q2. What is the datum of the current CRS for your map? (2 points)

Q3. What is the projection of the current CRS for your map? (2 points) We can change the CRS of the map to any CRS that we want. Notice that there are separate lists for “Geographic Coordinate Systems” and “Projected Coordinate Systems”.

Q4. What is the difference between a geographic coordinate system and a projected coordinate system? (10 points) **Step 4:** Select or search for “North Pole Azimuthal Equidistant” and double-click it or select it and click “Apply”. Inspect the result. We can also modify CRS definitions to create a custom map projection. For example, notice how Canada is rotated in this azimuthal projection because the default central meridian is 0° longitude. Open the map properties again, but this time right-click on “North Pole Azimuthal Equidistant” and select “Copy and Modify”. In the dialogue window that appears, find the value for “Central Meridian” and change it from 0 to -123, then click “Save”, then click “Apply”. Inspect the result. Notice that we have now rotated the map projection over Vancouver, Canada. Feel free to explore other projections, if you want, but before proceeding to the following steps, be sure to change your map CRS back to “WGS 1984 Web Mercator (Auxiliary Sphere)”.



What we just explored is that your ArcGIS Pro Map has its own CRS defined. This is an important concept because a map can only have one coordinate system, whether we are talking about a digital map in ArcGIS Pro or an historical paper map. It will be very important for you to understand that your ArcGIS Map can have a different CRS than the data that you want to map. ArcGIS Pro and other modern GIS software solve this issue by re-projecting your spatial data “on-the-fly” using the current CRS of your map. You see the illusion that everything is “mapped” in the same coordinate system, but in fact, your underlying data can be in completely different coordinate systems, even different datums. For our purposes in this lab, this is okay because we will only be visualizing the data as layers, but this issue becomes very problematic when we start to run data through tools and rely on the calculations of position, area, and perimeter/length of features, which can be very inaccurate when working between different CRS. Therefore, it is best practice to choose the CRS that is most appropriate for your study area and purpose and then change your map to that CRS.

Q5. What CRS would be most appropriate for your study area and why? Justify your reasoning. (15 points) As you will discover in the next task, spatial data are distributed in a wide variety of coordinate systems, and you are not always guaranteed the CRS that matches your project. So you may need to manually re-project your data before undertaking any analysis beyond simple visualization. Below is a table of three commonly used CRS in British Columbia and Canada:

Coordinate Reference System (CRS)	Uses
NAD 1983 UTM Zone 10N	Mapping within Universal Transverse Mercator (UTM) Zone 10 North around Vancouver

Coordinate Reference System (CRS)	Uses
NAD 1983 BC Environment Albers	Mapping across all of British Columbia
Canada Albers Equal Area Conic	Mapping across all of Canada

Task 2: Find your own spatial datasets

Spatial data are everywhere. For this task, you will begin to explore different kinds of spatial data portals and also the variety of data formats that you might come across. In later tasks, we will dive further into examining, assessing, and visualizing the datasets that you find.

The table below lists a few places to start your search:

Portal	URL	Focus / Notes
BC Data Catalogue	https://catalogue.data.gov.bc.ca/	Provincial datasets: forestry, land cover, cadastral, climate, transportation, natural resources
City of Vancouver Open Data Portal	https://opendata.vancouver.ca/	Municipal datasets: zoning, land use, parks, trees, utilities, transportation, planning
Government of Canada Geoportal	https://geo.ca/home/	Federal datasets: topography, land cover, soils, administrative boundaries
Lunaris	https://www.lunaris.ca/	National discovery service for research data across Canadian institutions
UBC Campus and Malcolm Knapp Research Forest Orthophotos and LiDAR	https://206-12-122-94.cloud.computecanada.ca/	UBC-hosted historical LiDAR and orthophotos of campus and Malcolm Knapp Research Forest

Portal	URL	Focus / Notes
UBC Geospatial Open Data	https://github.com/UBCGeodata/ubc-geospatial-opendata	UBC-hosted vector geospatial datasets, campus and research related
UBC Malcolm Knapp Research Forest and campus Vector Data	first-gis.ead.ubc.ca (only accessible by UBC students)	Password-protected PostgreSQL database mkrf that contains various spatial layers for Malcolm Knapp Research Forest and campus
Natural Resources Canada	https://maps.canada.ca/	Federal geospatial data: topographic maps, forests, geology, hydrography
Metro Vancouver Data Portal	https://open-data-portal-metrovancouver.hub.arcgis.com/	Regional planning: air quality, utilities, infrastructure, parks
Natural Earth	https://www.naturalearthdata.com/	Global vector and raster datasets at multiple scales
UN FAO GeoNetwork	https://data.apps.fao.org	Global agriculture, land cover, and environmental datasets

Step 1: Navigate to one or more of the portals linked above and search for three vector datasets that might be relevant to your project. You can select data of any themes that you are interested in, but you must follow these guidelines to receive credit:

- You must select one dataset each of points, lines, and polygons (i.e., 2D vector data only, and one of each type)
- You must only use the sources listed above
- You must properly cite the source of your dataset

If you wish to use data from the UBC PostgreSQL server for UBC Vancouver campus or Malcolm Knapp Research Forest, then consult Steps 4-6.

Step 2: Once you have found a dataset, download the file into your ArcGIS Pro project folder location that you noted from Task 1. Do not download files directly into your project's geodatabase. Most likely the files that you will find will be zipped (compressed), so you will also need to unzip (uncompress) the files into your project folder so that ArcGIS Pro can read them.

Pay attention to the metadata on the portal that describe what you are down-

loading. Metadata are data about data. They describe data and help to make data findable through portals and library indexes. For example, you might come across metadata that describes who the author of the data is, when the data were created/collected/updated, how large the files are and what formats they are, codes for understanding the attribute table, licensing, and more. Rather than provide detailed instructions here for each portal, just ask the instructor or TA if you have any doubts or questions about what you are looking at.

Step 3: For each dataset that you download, check the file type and then bring it into ArcGIS Pro using the instructions in the table below. For many file types, you can just drag and drop the file into ArcGIS Pro. You can also navigate to your project folder in the “Catalog” pane (expand “Folders”, expand “Lab1_Exploring_Spatial_Data”) and drag files from there or right-click files and select “Add To Project” (you may need to right-click the project folder and select “Refresh” to see recently added files from this method).

Format	File Extension(s)	Notes	Opening in ArcGIS Pro
Shapefile	.shp (+ .dbf, .shx, etc.)	Legacy, widely used, multiple files	“Add Data” > “Browse” > select .shp
GeoPackage	.gpkg	Modern OGC standard, single file, may contain many layers	“Add Data” > “Browse” > select .gpkg then choose layer
File Geodatabase	.gdb (folder)	ESRI native format, may contain many layers	“Add Data” > “Browse” > navigate into .gdb folder > select layer
GeoJSON	.geojson	Web-mapping format	“Add Data” > select .geojson
KML/KMZ	.kml, .kmz	Google Earth	“KML to Layer” > ArcGIS Pro converts to feature classes in a geodatabase and a layer file for styling
CSV with coordinates	.csv	Tabular data with X/Y columns	“XY Table to Point” > set fields

Steps 4-6 below are for optionally accessing data from the UBC PostgreSQL server for UBC Vancouver campus and Malcolm Knapp Research Forest. If you

are not using these data sources, then proceed directly to Step 7.

Step 4: Ensure that you are authenticated through UBC myVPN. If you are at UBC, use your dedicated ethernet port or connect via the ubcsecure wireless network. If you are away from campus, you will need to first connect to the UBC myVPN service using Cisco AnyConnect Secure Mobility Client. Only users authenticated with a Campus Wide Login (CWL) and connected to the UBC myVPN may access the UBC PostgreSQL server.

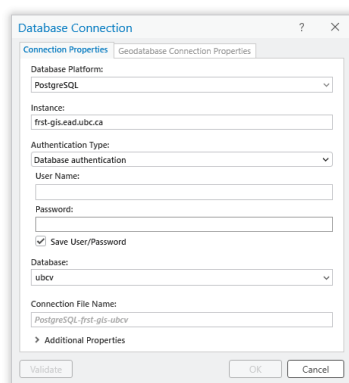
Step 5: From the top ribbon, navigate to the “Insert” tab, then click “Connections”, and from the drop-down menu, select “New Database Connection”. From the “Database Platform”, select “PostgreSQL”. Paste `FRST-GIS.ead.ubc.ca` into “Instance”, which is the hostname/address for the UBC PostgreSQL server.

You will need to get the student credential (username and password) from your instructor, which is different from your CWL, before proceeding. This is ordinarily provided during your registered lab session and will not be found in these assignment instructions.

Once you have correctly entered the correct credential, ArcGIS Pro will do a “soft” connection to the server to retrieve the available databases. From the “Database” drop-down menu, select `ubcv` for UBC Vancouver campus datasets or `mkrf` for Malcolm Knapp Research Forest datasets and then click “OK”.

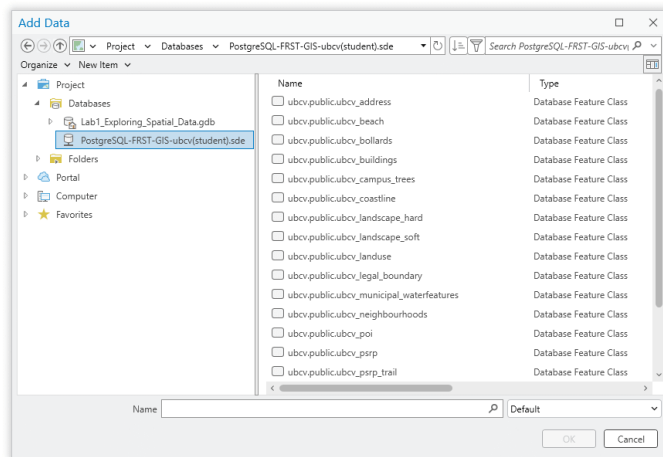
If you receive an error:

- Are you connected to myVPN or ubcsecure wifi network?
- Did you enter the correct server address provided above? (this is not case-sensitive)
- Did you enter the correct student credential, not you CWL? (this is case-sensitive)



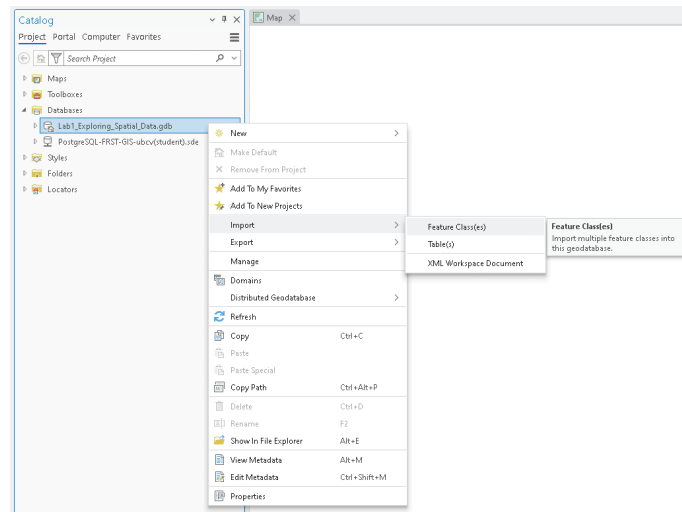
Step 6: From the top ribbon, navigate to the “Map” tab, then click “Add Data”. From the dialogue window that opens, expand “Databases”, and you will see your local project geodatabase that was created when you initiated the project

and the newly added postgres server. Click the postgres server connection to view the tables and then add a layer to your map. Note that your list of layers may be different if you connected to the `mkrf` database instead.



Note that ArcGIS Pro refers to vector data (i.e., points, lines, polygons) as a “feature class”. You will see this language throughout various tools in ArcGIS Pro (e.g., “Input Feature Class”, “Output Feature Class”).

Step 7: Import your data into your project’s geodatabase. This will vary depending on the file formats that you encounter, but generally you can right-click the geodatabase in the project folder, select “Import”, then select “Feature Class(es)”. In the “Feature Class to Geodatabase” tool that appears, select your three datasets from the drop-down menus (if you added them to your ArcGIS Pro Map already) or navigate to the location of the files in your project folder then click “Run”. When it completes, be sure to remove the old layers from your map (right-click the layer, “Remove”) and then add the three feature classes in your geodatabase to your map.



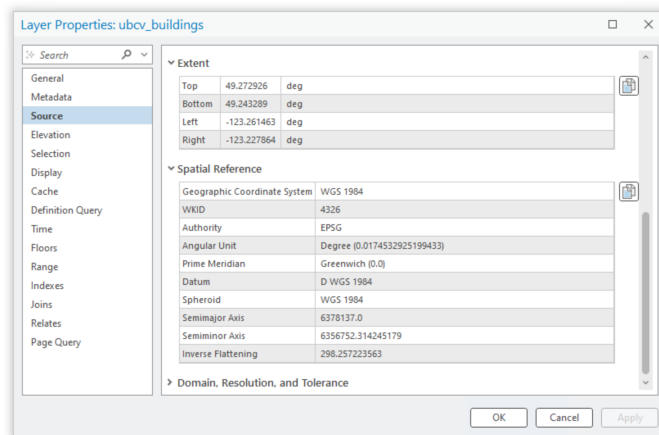
Up to this point, you have primarily been manipulating data directly through ArcGIS Pro, but you have not really changed much about your actual project. So there is no project state to really save. For example, if ArcGIS Pro crashed right now, you should still be able to find the feature classes that you imported into your project geodatabase. That said, it is good practice to save your project file moving forward as you work through this lab and later assignments.

Q6. What are the three datasets you chose and what sources did they come from? Provide the URL to the data source (or indicate use of the UBC PostgreSQL server) and report the file format (or PostgreSQL) that the data were provided in. (15 points)

Task 3: Inspect and Examine your Spatial Data in ArcGIS Pro

Ensure that you have added your three datasets as layers in your ArcGIS Pro Map.

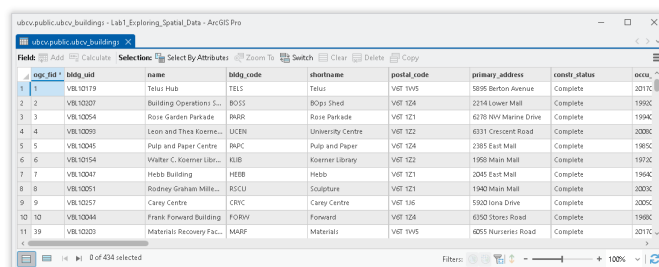
Step 1: Right-click on each layer in the “Contents” pane and select “Properties”. From here, you can view a lot of important metadata for your layers. Navigate to the “Source” tab from the list on the left and then expand “Extent” and “Spatial Reference”.



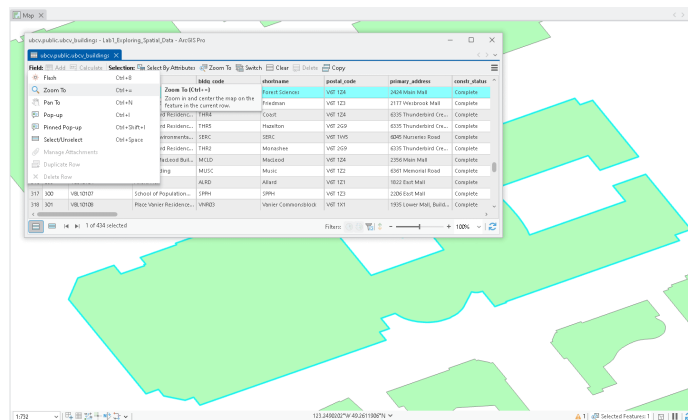
The extent represents the extreme spatial limits of the dataset, as if you drew a rectangle around all the features, what we call a bounding box. In the image above, the units of the extent for the buildings on UBC Vancouver campus are given in decimal degrees (“deg”). Below that, we see the CRS for these data is WGS 1984, a geographic coordinate system. In other words, these data are mapped purely in latitude/longitude values and are not projected. Remember that the CRS of our data can be different than the CRS of our ArcGIS Pro Map.

Q7. What is the CRS and extent for each of your three datasets? Paste the values from ArcGIS Pro and be sure to indicate which value corresponds to which dataset. (15 points) Close the layer properties.

Step 2: For each of your layers, right-click the layer in the “Contents” pane and select “Attribute Table”. Inspect the attributes. Are there any acronyms or codes that you are not familiar with? To decode those, you might need to refer to the metadata from the portal. From the top ribbon, click “Table”, then select “Fields”. Here we can see a quick overview of all the data types in the attribute table. Each field must contain values that are in the same data type.

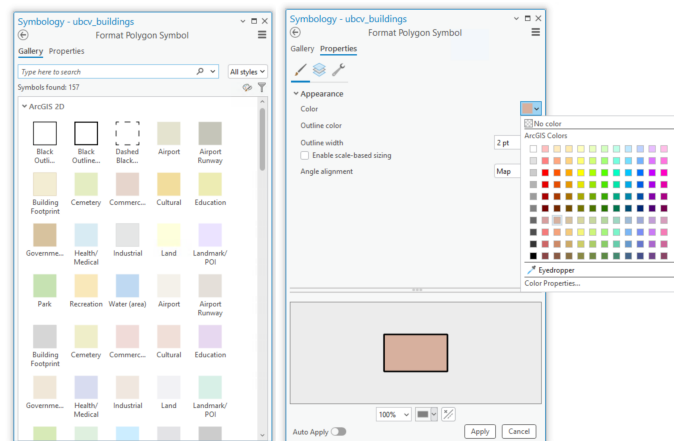


Q9. How many features are in each of your three datasets? (3 points)
From the attribute table, you can right-click on any row and select “Zoom To” and the map pan and zoom to that feature in the map.



In this last task, we will look at some simple ways to manipulate the symbology of your layers and produce and export a final map.

Step 1: Right-click one of your layers and select “Symbology”. Note that ArcGIS Pro randomly assigns a colour/symbology to a layer when it is added to the map. Very rarely do data have a pre-defined symbology that come with the data. The default symbology is typically a random colour that is applied to all features, this is called “Single Symbol”. For something like polygons, you can change the fill and the border by clicking the icon next to “Symbol”. From here, you can select any of a number of preset symbologies provided by ArcGIS Pro or you can select the “Properties” tab at the top and customize with whatever features you want.

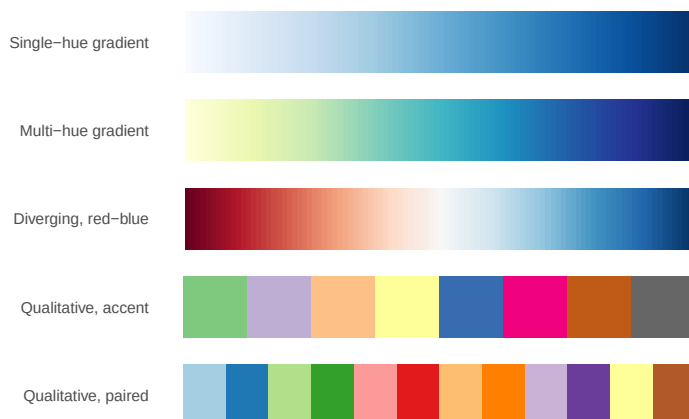


Experiment with the different symbology types for your layers and attributes until you arrive at something you like. Be sure to change the symbology of each of your layers. The sections below give some examples of how you can modify point, line, and polygon symbologies. You do not need to follow these sections exactly with the same data. They are only meant to give you some ideas for how you can symbolize your data and prepare your map.

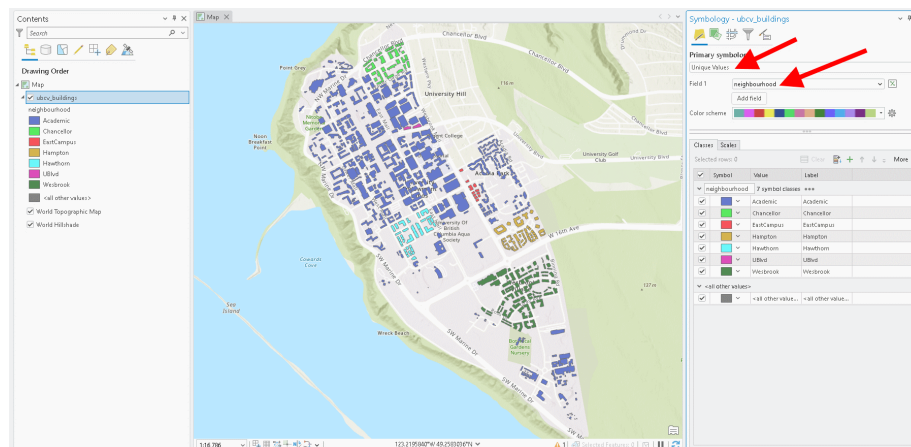
Polygon Symbologies

Polygons can be symbolized for their areas and/or their perimeters. Depending on the type of attribute that you want to symbolize (qualitative, diverging, sequential), you will choose a different symbology type and color palette. The figure below shows some examples of different types of colour palettes that are available in ArcGIS Pro.

ArcGIS Pro Palette Types

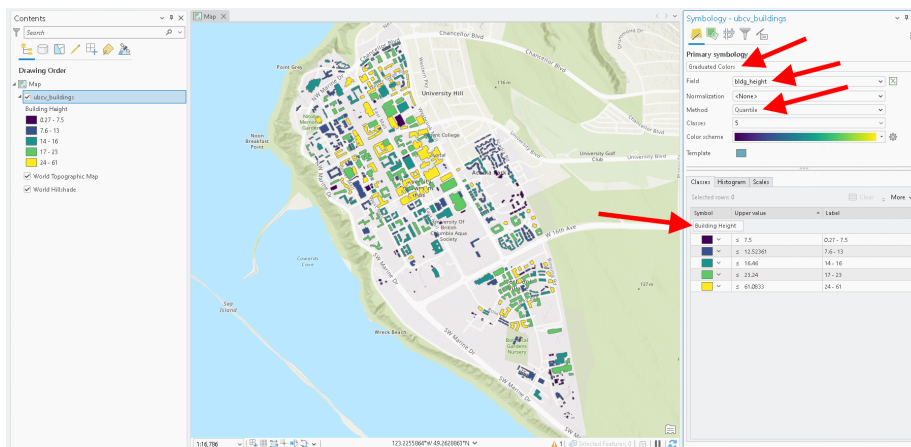


If the attribute that you want to symbolize is discrete or text-based, then you should select “Unique Values” for the “Primary Symbology”. Then, select the field from your attribute table that you want to use to symbolize different classes. This is a good choice for the “neighbourhood” field in the “ubcv_buildings” dataset shown below. When you choose “Unique Values” you should use a discrete color scheme. If your qualitative categories are completely distinct, you might choose a colour palette that uses accent colours.

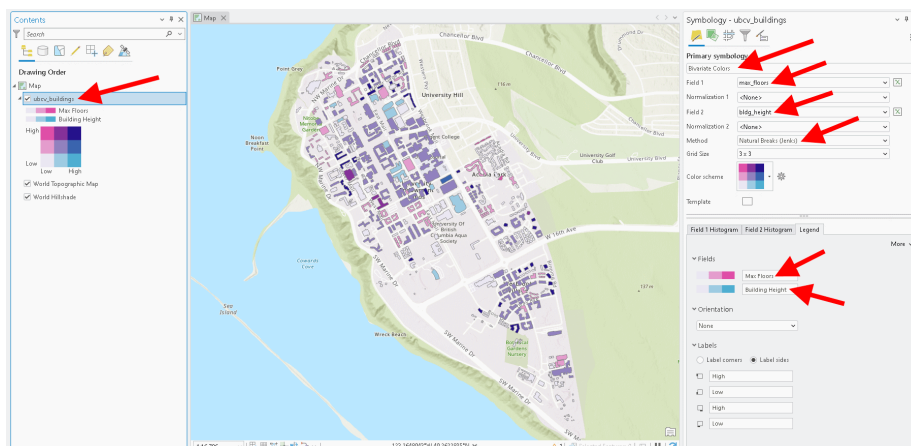


If your qualitative categories can be grouped in some way, like “deciduous” and “coniferous” are both types of forests, then you might choose light/dark paired colours so that all forests on your map are some shade of green. Similar logic can be applied to other classes like water, urban, agriculture, and so on.

For continuous attributes, you should choose either “Graduated Colors” or “Unclassed Colors” as the symbology type. Typically, you should use either the single hue or multi-hue gradients for these values. For example, you might see total precipitation displayed in a single hue while temperature is displayed in a multi-hue gradient. Diverging gradients should only be used for continuous attributes that have values that diverge from zero or if you want to emphasize values above or below a particular value. With “Graduated Colors”, you can classify the range of values into classes, which can sometimes improve the readability of the map, but it also creates more items for your legend (discussed later). You can choose the number of classes you want and the method for identifying the breakpoint values between classes. In the example shown below, the building height values are separated into five classes using quantiles. Note the small text box just above the symbols of the five classes on the right is where you enter the label as you want it to appear in the legend, “Building Height”.



In the case that you want to visualize two attributes at the same time from one layer, then you need to use a “Bivariate Colors” symbology type. This effectively combines two gradients, one for each attribute, into a grid matrix where the intersection of those gradients produces new hues. For example, in the image below max floors is shown in a single hue pink gradient and building height is shown in a single hue blue gradient. The mixture of these two gradients, produces a new set of blues and purples, depending on the value of both attributes for a given polygon. Where values for building height and max floors are both high, the polygon is coloured dark purple (upper right corner of the colour grid).

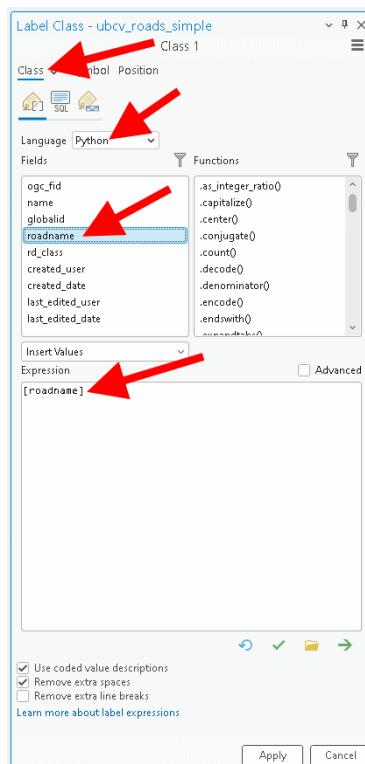


Line Symbolologies

Lines are often used to show route networks like roads or streams. Since the goal of such visualizations is to follow or navigate the path relative to other features in the map (points and polygons), in this section we will focus on labelling with a “Single Symbol” symbology type over the ESRI Imagery basemap.

When placing line features on a darker image basemap, we need to use a lighter colour such as white. To label these line features (or even points or polygons), right-click the layer in the Contents pane and select “Labels”. This will toggle on the labels for the layer. Next, right-click the layer again and select “Layer Properties...” to expose the labeling options.

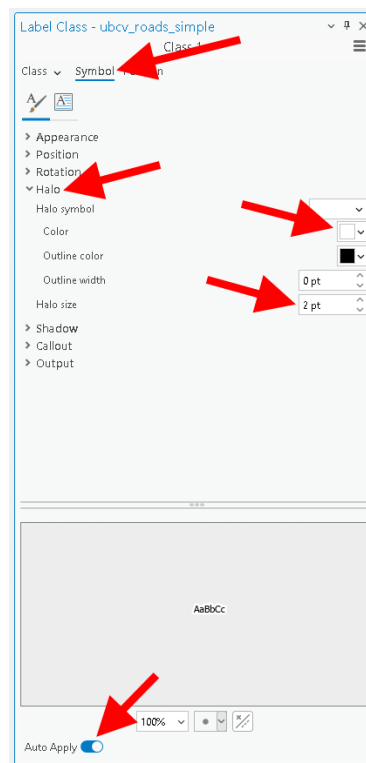
In the Class Label dialogue window that opens, you can specify the text label that will be placed next to the feature. There are a lot of options here, mainly because there are many ways that you can generate the text. Most likely, you just need to pick an attribute that contains the text for each line segment. This is exactly what is shown in the image below with the roadname attribute, using Python.



Notice that this accepts an “Expression”, which in the example above is simply `[roadname]` because this is the attribute that contains the text values that we want to use as labels for each feature. The expression is automatically iterated over all features. Below are some additional examples for how you might construct expressions to achieve different results.

Python Expression	Example Output	Description
"City Boundary"	City Boundary	A static text label
"Persons per sq km: " + str(round([POP2021] / [SHAPE_Area], 1))	Persons per sq km: 5356.7	Performs a calculation
"[CITY_NAME] + ", " + [PROV_ABBR]"	Vancouver, BC	Concatenates text

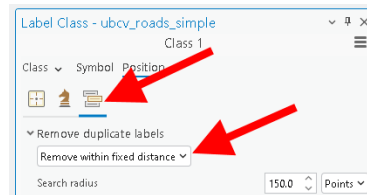
Next, we are going to add a white halo to the label so that the text is easy to see (along with the white line). Expand the "Halo" group under "Symbol" and change the colour to white. Adjust the size of the halo until you are satisfied. It is good to do this only when you are happy with the current extent and scale of the map since zooming in or out later will change the relative sizes of the text and crowding on the map.



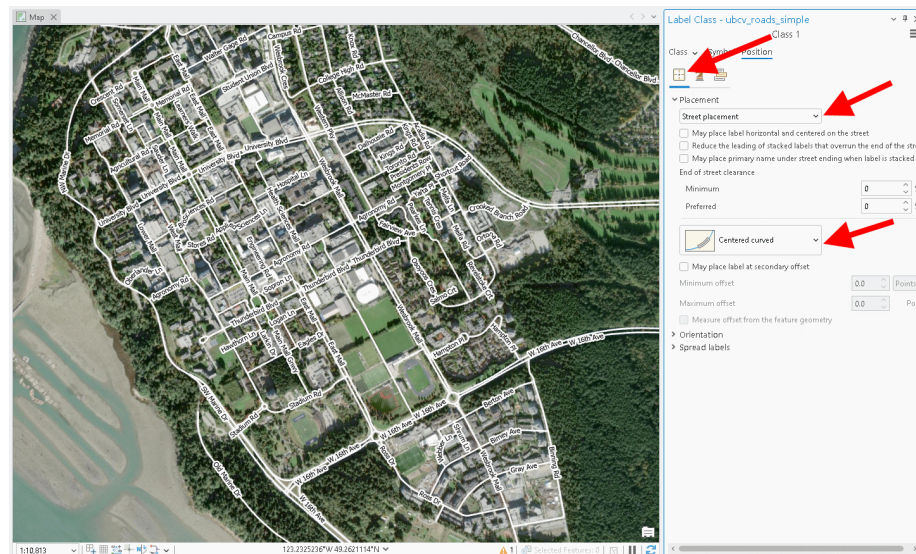
Positioning labels around features is a major challenge in cartography. There are so many factors that cause labels to collide. Generally, our goal is to position the labels so that the map remains readable and to do this mostly automatically since manual label placement is very time-consuming. Switch to the "Position" options and then select the chess knight icon and untoggle "Stack label". This will prevent long text labels from being stacked on two lines.



Next, click the label icon and under “Remove duplicate labels” select “Remove within fixed distance” from the drop-down menu. Change the search radius to “150” points. This prevents the same label from being drawn multiple times when a road is represented by multiple nearby (but disjoint) line segments that all have the same name.

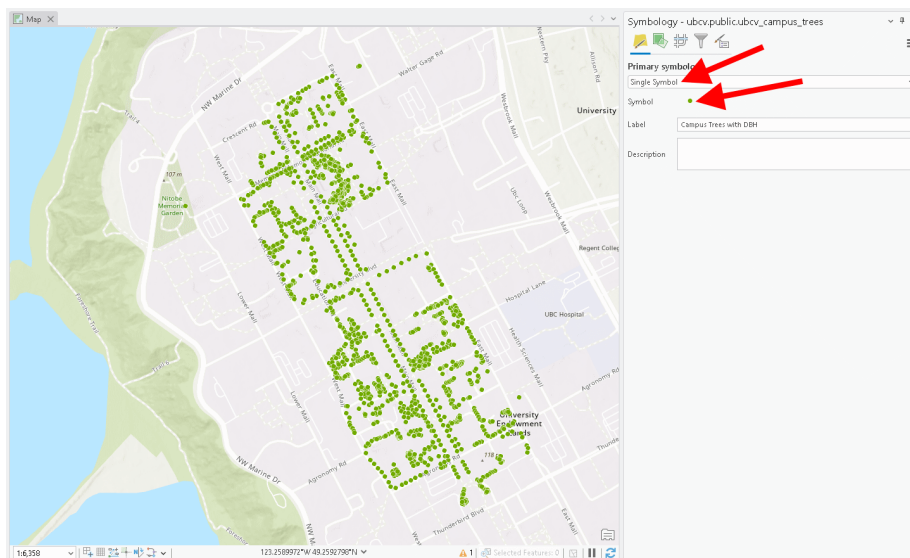


Finally, select the cross-hair icon, expand “Placement”, and choose “Street placement” from the drop-down menu. This option is similar to removing duplicate labels described above, but is specifically for multiple line segments that are connected and share the same label value, so ArcGIS Pro treats these as a single line segment for placing the label. Finally, you can choose where/how you want the label placed relative to the line feature. The image below shows an example of “Centered curved”, but you can also choose fixed offsets and these options will be slightly different if you are labelling points or polygons instead.

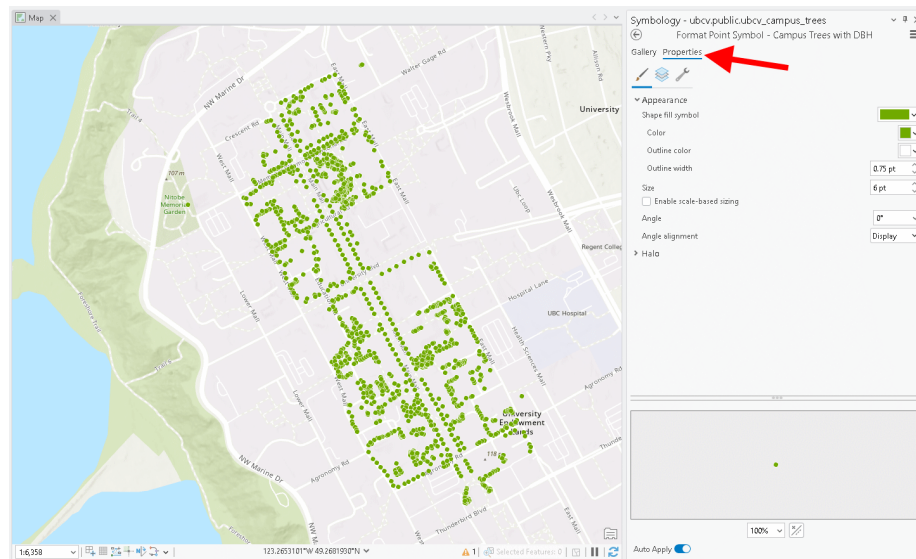


Point Symbologies

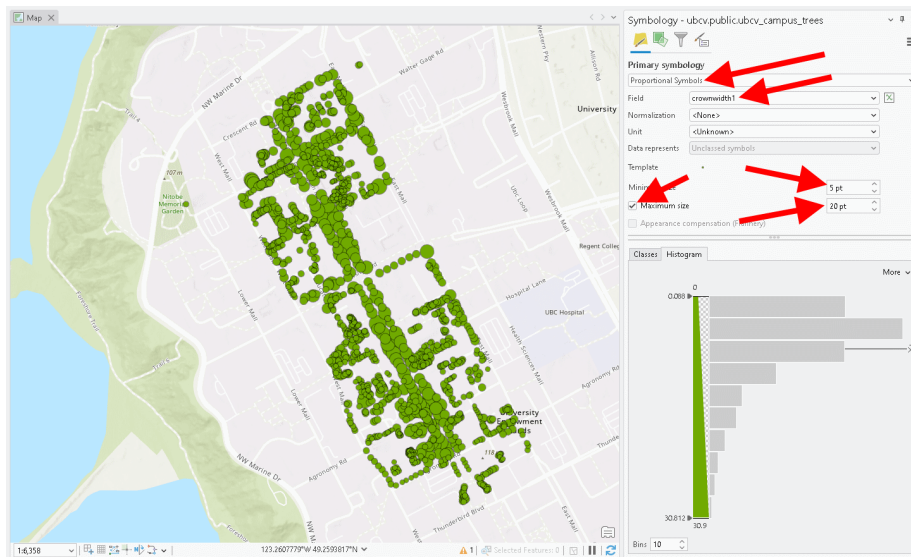
Points are spatially dimensionless, so their symbology requires different strategies compared to lines or polygons to achieve desired effects. For example, if you are just trying to convey location, then a simple “Single Symbol” symbology type will suffice as shown below using the `ubcv_campus_trees` dataset. Similar to polygons, you can assign colour palettes to attribute values using “Unique Values” or “Graduated Values” symbology types.



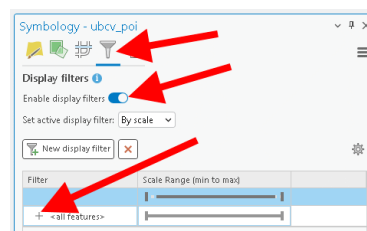
Clicking on the point symbol in the Symbology pane will allow you to choose from a “Gallery” of preset symbols and from there you can switch to “Properties” to customize your own point symbol.



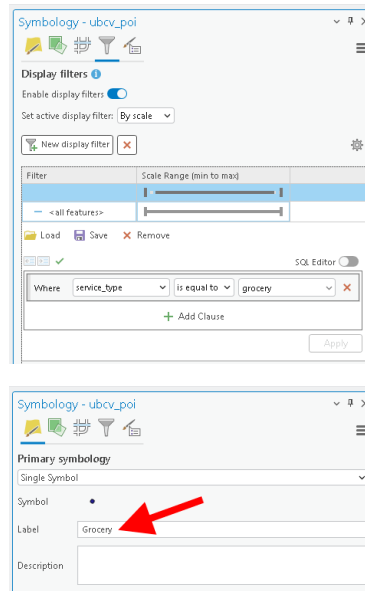
You may want to convey the value of an attribute by adjusting the size of the symbol, rather than the colour. This is a “Proportional Symbols” symbology type and it also gives the effect of exaggerating the area of a feature that otherwise does not have an area. The example below shows this effect using the “crownwidth1” attribute of the campus trees. The overlapping features produce an illusion of continuous canopy cover, which has a different purpose than just showing the location of the trees. Adjusting the minimum and maximum point sizes can increase or decrease the amount of overlap. Compare the map below with the one above.



Points, like lines and polygons, can be filtered so that your symbology only applies to features that match some criteria. You can access these options by selecting the funnel symbol at the top of the Symbology pane, toggle on “Enable display filters”, then click the “+” next to “”.

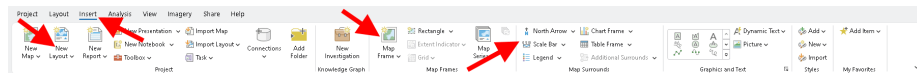


An option will expand where you can “Add Clause” and create an SQL statement to achieve your desired effect. This also works with lines and polygons, and you also have the option to change whether features are visible at different scales with the slider bar.



When you are satisfied with your layer symbologies, save your project because your project file stores the symbologies for your map layers. Now you can enter the map layout mode to begin the final layout and export process.

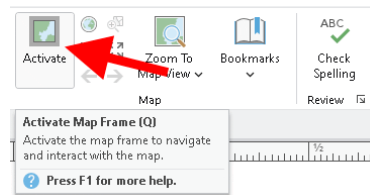
Step 3: From the top ribbon, select “Insert”, and then “New Layout”. We are not actually printing the map, so you can select a regular letter size layout in either portrait or landscape mode. This will open a new blank Layout tab in your project and expose a bunch of new options specific to exporting a professional map.



Step 4: From the “Layout” toolbar, select “Add Guides”, then add “Both” vertical and horizontal guides “Offset from edge” and use 0.25” as the margin. Now, when you place elements onto the layout, you can snap them to these guides.

Step 5: Add your ArcGIS Pro Map to your layout by selecting “Map Frame” from the top ribbon under the “Insert” tab. From there, you can select any Maps that you have created in your project. When you select your Map, your cursor will switch to a cross hairs and you just left-click and drag a rectangle where you want the map to appear on the layout. Once placed, you can drag the corners to resize the Map.

If you need to zoom in/out of the map without changing its position in the layout, then select “Activate” from the “Insert” toolbar or by right-clicking the map in the Contents pane.

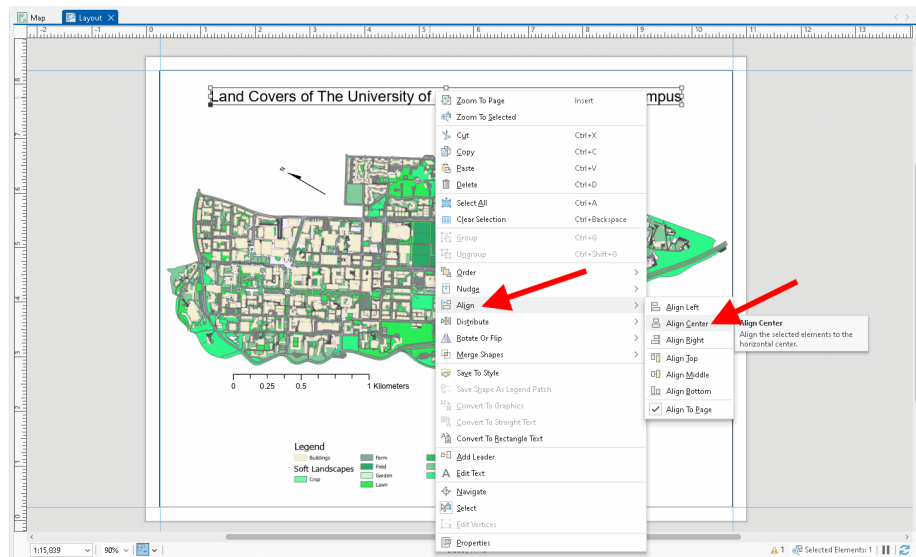


If you need to resize or reposition the map frame in the layout without zooming in/out or panning the map itself, then select “Close Activation” under the “Layout” toolbar or by right-clicking the map in the Contents pane.

Step 6: You can reorient your map so that it better fits the area of your page. In the example below, we have rotated the map of UBC campus by 60 degrees to give a more natural looking arrangement for the other elements. To do this, right-click the map in the Contents pane, select “Properties”, and then change rotation to your desired value.

Step 7: Add a North Arrow, Scale Bar, and Legend from the top ribbon under the “Insert” toolbar. Pick your favourite styles for these features and experiment with the large number of customization options until you achieve something you like. You can directly rename the layers in your “Contents” pane to update how they appear in the legend. You may need to go back to your individual layer symbologies to adjust how the values are grouped and displayed.

Step 8: Add a title to your map by select the text icon in the “Graphics and Text” grouping on the top ribbon under the “Insert” toolbar. Click to place the text and then type what you want. Right-click the element and select “Properties” to adjust font size, style, etc. Right-click the element again, select “Align”, then select whichever way you would like to align the text relative to the page or simply drag the element and use your guide lines to snap it in a location relative to the margins that we placed earlier.



Step 9: When you are ready to export your map, navigate to the “Share” tab from the top ribbon, select “Export Layout”, and choose “Georeferenced PDF”. You can accept most of the default settings in the dialogue window that opens, just note that the default output location is your project folder.

At this point, it will become apparent how visible features are based on your symbology choices. For example, heavy line weights or large point symbols can make it difficult to see other labels or nearby features, which may be important if you were using this map to navigate in the field. Resize your elements, adjust text label and symbol sizes, choose a larger page size for your layout, and remove compression to improve your desired output, at the expense of a larger file size. Here we chose a georeferenced PDF firstly because PDF’s are optimal for storing and displaying vector graphics and secondly because *georeferenced* PDF’s can be imported into other other mapping apps for use with field work.

Map 1. Upload your final map that shows your three selected datasets. At least one dataset must be symbolized by a field in the attribute table. You can pick the colors/symbols to use for each layer. Layers that appear to use the default random color/weight assignment by ArcGIS Pro will not be accepted for credit. You must show that you manipulated the symbology for each layer. Your map must also contain a title, north arrow, scale bar, and a legend that contains the symbologies of all the layers. (25 points)

Summary

In this lab, you learned how to find, evaluate, and visualize vector spatial data for a study area that matters to you. You searched multiple open data portals, chose relevant vector spatial data, and practiced importing external files into your project geodatabase. You inspected metadata to identify each dataset's coordinate reference system, extent, and field data types; distinguished geographic vs. projected coordinate systems; and justified an appropriate CRS for your study area. You also examined attribute tables to interpret codes and data types, then customized symbology (single symbol, graduated colors, etc.) so each dataset communicates something meaningful. Finally, you assembled a clean map layout and exported a finished map showing all three datasets. You will likely repeat nearly all of these tasks for almost every GIS project you undertake in the future, and we will continue to build on this foundation in later labs.

Return to the **Deliverables** section to check off everything you need to submit for credit in the course management system.

Chapter 2

Collecting and processing GNSS data

Written by Paul D. Pickell and Amy Blood

Lab Overview

Oftentimes, you will need to collect and display your own data. There are many phone applications to collect your own GNSS data. In this lab, you will plan your own GNSS collection, collect data, and process the data.

Learning Objectives

- Plan for field data collection
 - Use AvenzaMaps to collect GNSS readings
 - Evaluate the accuracy and precision of GNSS coordinates
-

Deliverables

Responses to the questions posed throughout the lab on the course management system. (57 points)

Screenshots of your Trimble GNSS Planning tool charts (your settings must be visible in all the screenshots): Number of Satellites, DOPs, Iono Information, Sky Plot. (8 points)

Map of the “True” and “Observed” points. (25 points)

Table of accuracy and precision calculated values. (10 points)

Task 1: Preparing for GNSS Data Collection

Field data collection is an important skill set to learn and practice. In this task, you will plan the collection of your GNSS data in a public park. The task for collecting GNSS data in a public park has nearly the same risks as if you collected the data on campus and you are expected to take similar precautions. This section is meant to inform you of the likely hazards and how to stay safe.

Planning Your Data Collection Safety starts at the planning stage. When planning which park you will visit, consider somewhere nearby that is easily accessible by transit or walking. Avoid going somewhere that is unfamiliar to you. Ensure that you are visiting a public park and stay off of private or restricted property. You should only visit the park during daylight hours and plan to go out in sunny weather only. Check government websites for any recent animal sightings before you commit to going to your desired park. Do not plan to collect your GNSS data near bodies of water (e.g., oceans, lakes, ponds, rivers, streams). Do not plan to collect your GNSS data near cliffs or on steep terrain. Finally, ensure that you will have good cell phone coverage in case of an emergency.

Likely Hazards At all times, you must be aware of possible hazards both overhead and underfoot. The main overhead hazard in a public park is going to be trees and falling branches. Do not collect your GNSS data during windy or stormy weather, which may cause tree branches to fall. Wet and slippery surfaces, steep angles, holes, logs, debris, and loose soil all pose fall hazards. Many of these hazards can be avoided with careful planning before you even step outside. Speaking of stepping, make sure you wear appropriate footwear, closed toed shoes are best for this work. Fauna are natural inhabitants of parks. Do not visit parks with recent sightings for large, predatory fauna such as bears or cougars. Even urban parks like Stanley Park in downtown Vancouver are known to have coyotes who have attacked people. As well, avoid areas with hazardous flora that may be thorny or poisonous. You may need to cross or transit streets to reach your park. Always follow local traffic laws and look for moving vehicles in and around your park. Always use designated crosswalks and do not look at your phone when walking near stopped or moving vehicles.

Do not collect your GNSS data while walking and looking at your phone. Always be aware of your surroundings.

Important: If you feel uncomfortable undertaking this task, please contact the instructor for alternative arrangements or accommodations for this particular assignment

For this lab, you will decide on your own study area through your own knowledge

and simple remote sensing.

Your study area must:

- Be somewhere that you can legally and safely visit
- Be a park or a greenspace
- Have at least a portion of its ground visible using aerial imagery (e.g. Google Earth or an ArcGIS Pro basemap)

Q1. Describe your study area. Where is it located? What features are around? Why did you choose this location? (5 points) **Step 1:** Open ArcGIS Pro and turn on an imagery basemap (“Map”, “Basemap”, and “Imagery”). Navigate to your proposed study area. You can do this by (a) dragging around your map, (b) putting coordinates into Go To XY, (c) typing in an address or park name into the Locate tool.

Before physically visiting your study area, you will need to verify that there are usable reference points in your chosen park. The reference points should be immovable, viewable from the sky and ground, and not tall (not trees or buildings). Additionally, you will be walking to your points – so they should be somewhere safely accessible and not too close or too far from each other.

Step 2: Find at least five possible reference points (all at least 10 m from each other). You will create a new feature class and populate it with these reference points. In “Geoprocessing”, find the “Create Feature Class” tool by searching for “Create Feature Class”. Save it to your ArcGIS Pro project geodatabase and give it a meaningful name (for this lab, it will be called “refpoints”). For the other options:

- **Geometry Type:** Point
- **Has M:** No
- **Has Z:** No
- **Coordinate System:** –Local UTM–
- Click “Run”

Your new feature class should now be in the “Contents” Pane, but it has no points.

Q2. Describe each of your five reference points. What features do they represent? (5 points) Choose “Edit” from the top ribbon, then select “Create”. On the right, a “Create Features” pane will appear. In there, click refpoints, then select “Create a Point Feature”. Zoom in closely on the basemap before creating point features so that the points are more accurate. Click on (at least) four reference points, then select “Save” from the Edit ribbon. If you need to Move a reference point, you can select Move from the Edit Tools.

Open the attribute table of refpoints and add three new columns: Name, East, North. Populate the columns for each reference point: Name – a unique name. East – Easting. North – Northing. Easting and Northing can be populated using

“Calculate Geometry.” For more reliable measurements in your precision and accuracy assessment (Task 3), use your local UTM for the geometry. Be sure to save your edits.

Step 3: You will be exporting a georeferenced map of your study area. Zoom in (or out) to an extent that contains at least the entire park.

Using the top ribbon, select Share , then Export Map. A pane will pop up on the right-hand side of your screen. The file type is PDF, and make sure that it is saved somewhere you will be able to find it. Toward the bottom of the pane, make sure that the box under “PDF Settings” that says “Export georeference information” is selected. Then select “Export”. This will be used when you are collecting data with Avenza.

Step 4: Now that you have established a study area, you need to decide when you will be collecting your data. There are times in the day where you are more likely to get accurate GNSS readings. Find the coordinates of your study area so that you can use them in the Trimble GNSS Planning Online website (<http://www.gnssplanning.com/>).

Go to the Trimble GNSS Planning Online website and enter your study area’s coordinates (one set of coordinates from anywhere within the study area) and elevation. If your coordinates are not in the proper units, either convert them to the proper units. Choose the day that you plan to visit your field site and collect data.

After you have input your information into Settings, click “Apply” and look through the other tabs (Satellite Library, Charts, Sky Plot, World View). Using the information on the website, determine the ideal time to collect your GNSS points. Justify your chosen time using the data on the website (this justification should be included on your report, along with screenshots from the webpage).

Screenshot 1. Upload a screenshot of the Number of Satellites chart from the Trimble GNSS Planning Tool. The settings that you used must be visible in the screenshot. (2 points)

Screenshot 2. Upload a screenshot of the DOPs chart from the Trimble GNSS Planning Tool. The settings that you used must be visible in the screenshot. (2 points)

Screenshot 3. Upload a screenshot of the Iono Information chart from the Trimble GNSS Planning Tool. The settings that you used must be visible in the screenshot. (2 points)

Screenshot 4. Upload a screenshot of the Sky Plot from the Trimble GNSS Planning Tool. The settings that you used must be visible in the screenshot. (2 points)

Q3. Interpret the Trimble GNSS Planning Tool charts and sky plot with 1-2 sentences each. When is the ideal time to collect data at your study area? When did you actually collect data at your study area? (15 points)

Task 2: Collect GNSS Data

You will need either an Android or iOS smartphone capable of installing the Avenza Maps App. If you do not have an Android or iOS smartphone, then please contact the instructor for alternative arrangements or accommodations

Step 1: To start you will need to install Avenza Maps App on your phone. There is an Android version as well as an iOS version so just go to your app store and download it. It's free!

Step 2: Once you have the app installed on your phone, you need to sign up for a free account. If you have concerns about your privacy, you can use a pseudonym and your student UBC e-mail address. Avenza Maps is a widely used mobile mapping application, so chances are you might continue to use it in the future for other projects or hobbies. If you want to ensure future access to your Avenza Maps account and assets, then choose an e-mail account that you will continue to have access to (i.e., not your UBC student e-mail).

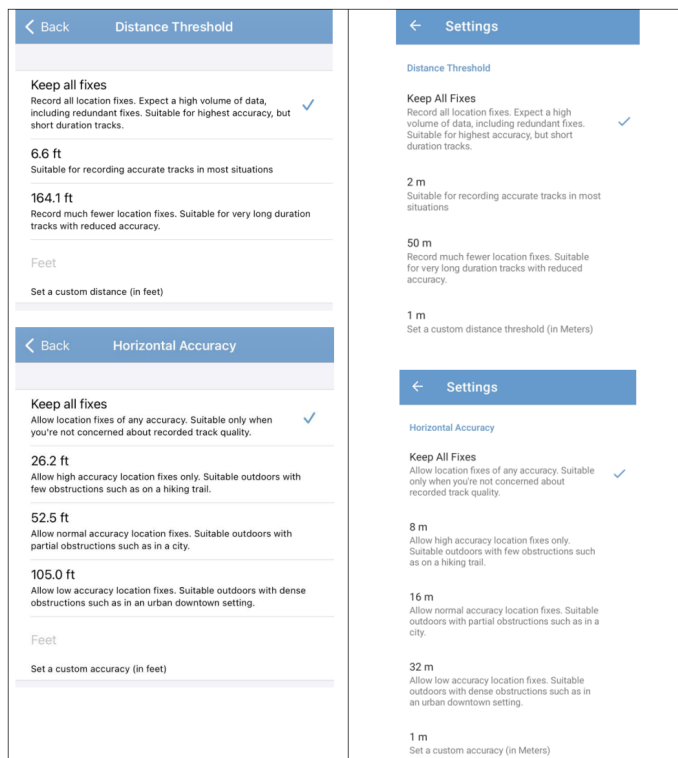
Step 3: Now that you are registered and logged in, you will change some settings in the Avenza Maps app. Navigate to the Settings in the bottom right corner (for iOS) and in the top right corner (for Android). Scroll down and click "GPS settings". Use the following settings:

- **Location Source:** "Use Only Internal Sources". This ensures that we are only using the phone's internal GNSS receiver.
- **Horizontal Accuracy:** "Keep all fixes"
- **Distance Thresholds:** "Keep all fixes"
- **Time Threshold:** "Keep all fixes"
- **Ignore Suspicious Fixes:** Toggled Off

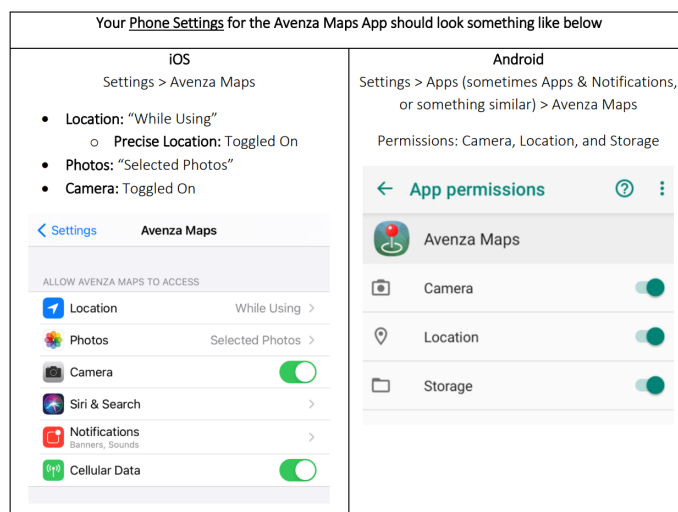
Next, we will set the units to metric. In Settings, navigate to "Units of Measurement" and select "Metric" at the top. Under "Coordinates", ensure that the "Format" is set to a workable coordinate system.

Your Avenza Maps App Settings should look something like below

iOS	Android
Location Source Use Any Source Both external GPS sources and device based location services will be used for location and tracking. Use Only External (Trimble) Sources Only External Trimble GPS Device sources will be used for location and tracking Use Only Internal Sources ✓ Only Device Location services will be used for location and tracking.	Settings Location Source Use Any Source Both mock locations and device based location services will be used for location and tracking. Use Only External Sources Only Mock location providers will be used for location and tracking. If mock locations are not sent by the provider, location and tracking tools will not update. Useful if you have an external GPS that you wish to use exclusively, for example, Trimble devices Use Only Internal Sources ✓ Only Device Location services will be used for location and tracking.
Time Threshold Keep all fixes ✓ Record all location fixes. Expect a high volume of data, including redundant fixes. Suitable for highest accuracy, but short duration tracks. 2 seconds Suitable for recording accurate tracks in most situations 30 seconds Record much fewer location fixes. Suitable for very long duration tracks with reduced accuracy. Seconds Set a custom time (in seconds)	Settings Time Threshold Keep All Fixes ✓ Record all location fixes. Expect a high volume of data, including redundant fixes. Suitable for highest accuracy, but short duration tracks. 2.0 seconds Suitable for recording accurate tracks in most situations 30.0 seconds Record much fewer location fixes. Suitable for very long duration tracks with reduced accuracy. 1 second Set a custom time threshold (in seconds)



Step 4: Ensure that the Avenza Maps App is allowed to access your location, camera, and photos in your *phone's settings*.



Important: Personal location is very sensitive information. You should not

undertake this exercise at your home or share the coordinates or screenshots of your home location in Avenza Maps with anyone, including the instructor, TA, or other students in this course. Treat personal location information as any other private sensitive information and handle with respect and care.

Step 5: Avenza Maps requires you to load a map before being able to use the location service of your phone. You will import the georeferenced PDF map that you created in Task 1. There are several ways to do this in Avenza Maps, but probably the easiest is to download the PDF to your phone and then import directly from your phone. *Be aware that map size can be large and will consume cellular data if you are not connected to a wifi network.*

If you ran into problems creating your georeferenced PDF, then you can use the provided georeferenced PDF on Canvas for the world. In the end, the actual map that you use is not important for collecting your GNSS data, but Avenza Maps requires a georeferenced map be loaded in the app.

Step 6: Travel to your study area! Open the Avenza app, and select the Georeferenced PDF under “My Maps.” Go to one of the reference points you selected in ArcGIS. On the bottom right of the app, there is a button that will bring the map to your current area. Press that button, then press the placemark button directly to the right of it. Create a placemark and name it something meaningful (for example: placemark 1, location 1). Stay exactly where you are, and create a placemark every 10 seconds for the next minute by pressing the location button then the placemark. You should have a total of 6 placemarks. Take a picture of the area you are trying to geographically capture. Go to two more of your designated locations and do the same procedure (create 6 placemarks, each 10 seconds apart, and take a picture).

Step 7: Now you can return back to your workstation (after taking a nice walk!). In Avenza, go back to “My Maps,” then select Layers (at the bottom). Select the layer you created, then the icon with three lines at the bottom right. Choose export layers. Export the layer as a KML by pressing the EXPORT option at the top right. A list of export applications should pop up—choose whichever one you would prefer.

Step 8: Find your exported file and bring it into ArcGIS Pro. You can drag and drop it from the Catalog Pane into the main map. From there, you should export the kml file to layer using the tool “KML to Layer”.

Screenshot 5. Upload an image from one of your reference points. (2 points)

Map 1. Upload a map showing the True and Observed points. (25 points)

Task 3: Assess Precision, Accuracy, and Possible Errors

For this lab, we will consider the initial ArcGIS-created reference points as the “true” values and perform an accuracy assessment with that in mind.

Step 1: Visually compare the Avenza-created Placemarks (called “Observed values” for the rest of the lab) with each other and ArcGIS reference points. How close/far are they from one another? You can use the measure tool to add values to this visual assessment. When you are comparing the observed values to each other, you are assessing precision, when you are comparing them to the “True” locations, you are assessing accuracy.

Step 2: Quantitatively determine the accuracy of the points. Add two columns in the Observed attribute table and populate them with the east and north values for each point. You should have the True values from your reference point attribute table. Export both tables and bring them into excel.

Calculate the horizontal accuracy for each placemark using the following equation:

$$\sigma_{H_{acc}} = \sqrt{(\bar{E} - E_{true})^2 + (\bar{N} - N_{true})^2}$$

Excel is a convenient place to use this equation. Breaking down the equation – get the mean observed East value, then subtract the “True” East value. Square this difference. Do the same for North. Add the north and east results, then take the square root of them. In excel, you can do this with the equation:

```
=SQRT((average(observed east values)-true east value)^2+((average(observed north values)-true north value)^2))
```

The resulting value represents how accurate the GNSS measurements were – a lower value represents greater accuracy. Is there a noticeable difference in accuracy between your placemarks? Why do you think this is?

To assess the precision, you will use the following equation:

$$\sigma_{H_{pre}} = \sqrt{\sigma_E^2 + \sigma_N^2}$$

Calculate the standard deviation of the observed East and North values, square them, then add them and take the square root of the sum. In excel, you can do this with the equation:

```
=SQRT((STDEV(observed east values)^2 + STDEV(observed values)^2))
```

The resulting value represents how precise the GNSS measurements were – a lower value represents greater accuracy. Is there a noticeable difference in precision between your placemarks? Why do you think this is?

Table 1. Submit a table that shows the accuracy and precision calculated values for each point. (10 points)

Q4. Interpret the map and table. (5 points)

Q5. Discuss the potential errors in your data collection and describe what may have affected accuracy and precision. Integrate your site observations with the information from the Trimble GNSS Planning tool. Finally, reflect on the validity of the “True” points that you derived from the basemap imagery. (25 points)

Summary

In this lab, you learned how to plan, collect, and process GNSS data using a combination of ArcGIS Pro, Trimble GNSS Planning tools, and the Avenza Maps mobile application. Planning is a critical part of the GNSS collection process for both safety concerns and also to acquire the best satellite signals. Using the Trimble Planning Tool, you evaluated satellite availability, DOPs, and ionospheric conditions to select an ideal collection time. In the field, you practiced recording GNSS coordinates at reference points with Avenza Maps, producing multiple observations to assess variability. Back in the lab, you compared these “observed” values with the ArcGIS “true” reference points to calculate both precision and accuracy. This exercise highlighted the limitations of phone-based GNSS receivers, the effects of environmental conditions on data quality, and the importance of systematic planning and error assessment in field data collection. Overall, this lab showed you not only the technical workflow for GNSS data collection and analysis but also how to critically evaluate results by integrating field observations with planning tools and quantitative measures of accuracy and precision.

Return to the **Deliverables** section to check off everything you need to submit for credit in the course management system.

Chapter 3

Cartographic Modelling with Forest Inventories

Written by Sadie Russell

Lab Overview

Old growth forests are an important feature found in many types of forested ecosystems. They are characterized not only by their age but also by their complex composition and structure. (LePage & Banner, 2014) Old growth stands boast the highest rates of biodiversity and provide critical habitat for rare, threatened, or endangered species. Additionally, they contribute to soil fertility by supporting intricate soil structures formed by micro-organisms, fungi, and plants.

It is generally assumed that the structural complexity desired in a forest develops over time. As such, a stand is designated as old growth once it has reached a certain age without experiencing major disturbance. The length of time required for this designation varies by Biogeoclimatic Ecosystem Classification (BEC) zone.

Old growth forests are also highly valued for timber production due to their superior qualities. (Canadian Institute of Forestry, 2022) These trees have had the opportunity to grow slowly over long periods, resulting in denser wood grain and stronger lumber. Furthermore, their tall, straight trunks are ideal for producing a wide range of wood products.

To balance the benefits of both logging and preserving old growth forests, frameworks and guidelines are put in place to manage Old Growth sites. In BC, Old Growth Forests are assessed using the Cumulative Effects Framework (CEF). This framework was put in place to provide guidelines on assessing the effects

of human activities and natural processes on the provinces forested ecosystems. The provincial old growth and mature-plus-old targets are outlined in the Biodiversity Guidebook (Province of B.C., 1995).

Vancouver Island, British Columbia, has been an area of highly contentious Old Growth logging. Conflicting interests between logging companies, First Nations, Provincial and Federal Governments, and the general public came into the spotlight during the Fairy Creek protests of 2020. Over two years of protest, more than 1,000 protesters were arrested at the site (CBC, 2022). The events at Fairy Creek, and many others like it, have brought the issues of Old Growth conservation to the forefront of timber production relations.

Learning Objectives

By the end of this lab, you will be able to:

- Prepare and clean a forest inventory dataset and related layers for analysis
 - Perform attribute/select-by-expression filtering to isolate old-growth candidate polygons
 - Calculate old-growth area and percentages by polygon using field calculations
 - Produce a cartographic summary (map + table) that communicates old-growth distribution
 - Reflect on limitations and assumptions of VRI-derived old-growth estimates
-

Deliverables

Responses to the questions posed throughout the lab on the course management system (60 points)

A table with fields for total area, old-growth area, and percentage old growth (10 points)

A screenshot of the python expression created for Task 2 (10 points)

A map showing the study area and the distribution of difference in calculated old-growth percentage and targets, by polygon (20 points)

Data

All data for this lab are accessible via the UBC PostgreSQL server. Instructions for connecting to the server are given in the tasks below. These are very large datasets that come from the BC Data Catalogue and are described in the table below. Accessing these data from the UBC PostgreSQL server allows us to extract only the data we need for this lab, which is the extent of Vancouver Island.

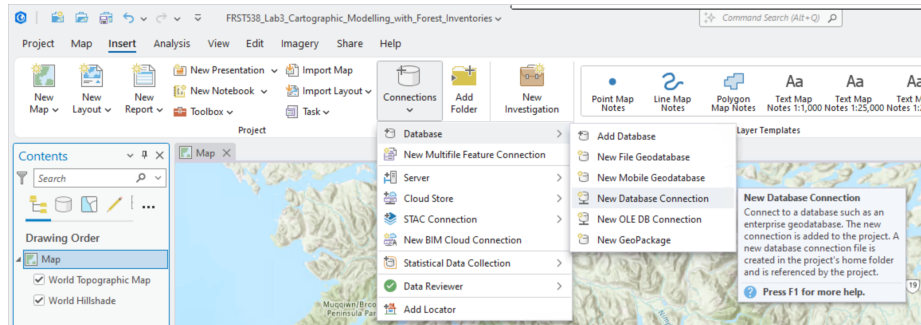
Layer Name	File Name	Reference Link
Vegetation Resource Inventory 2025	<code>vancouver_island_vri</code>	https://catalogue.data.gov.bc.ca/dataset/vri-2025-forest-vegetation-composite-layer-1-l1
Land Designations for Forest Management	<code>vancouver_island_own</code>	https://catalogue.data.gov.bc.ca/dataset/4574f131-470e-407c-bbcc-5b19d9adcc1b
Human Disturbance – 2025	<code>vancouver_island_human_disturbance</code>	https://catalogue.data.gov.bc.ca/dataset/human-disturbance-2025
Landscape Units of British Columbia - Current	<code>vancouver_island_landscapeunits</code>	https://catalogue.data.gov.bc.ca/dataset/landscape-units-of-british-columbia-current

Task 1: Query and export data from UBC PostgreSQL server

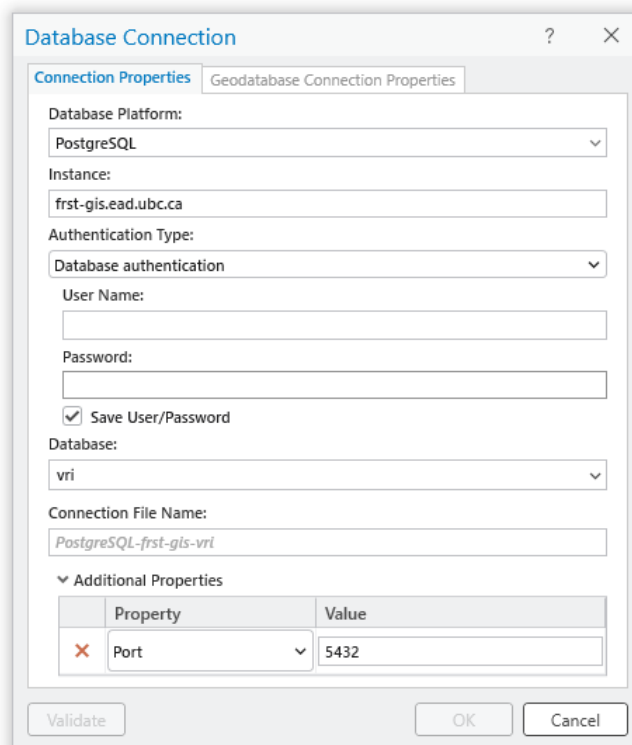
Step 1: Ensure that you are authenticated through UBC myVPN. If you are at UBC, use your dedicated ethernet port or connect via the ubcsecure wireless network. If you are away from campus, you will need to first connect to the UBC myVPN service using Cisco AnyConnect Secure Mobility Client. Only authenticated users with a Campus Wide Login (CWL) who are connected to the UBC myVPN may access the UBC PostgreSQL server. However, if you are not a UBC student, then you can just download the data directly from the BC Data Catalogue.

Step 2: Start ArcGIS Pro and create a new Map project.

Step 3: From the top ribbon, navigate to the “Insert” tab, then click “Connections”, and from the drop-down menu, select “New Database Connection”.



You can name it “UBC PostgreSQL Server” and for the host enter “FRST-GIS.ead.ubc.ca”. The username and password will be provided to students enrolled in the course and it is not your CWL. Once you have correctly entered the correct credentials, ArcGIS Pro will do a “soft” connection to the server to retrieve the available databases. From the “Database” drop-down menu, select “vri” and then click “OK”.



The database connection has now been added to your ArcGIS Pro project, and you can view the tables inside the database from the Catalog Pane (open the “View” tab, open the “Catalog Pane”, expand “Databases”, expand “PostgreSQL-FRST-GIS-vri(student)”, and view the tables in the `vri` database). You can drag and drop any of the layers of the database directly into your map to visualize them if you want. Though be aware that these are large layers and they may take a while to draw on your map. We do not actually need all of these data, so instead, we are going to use the power of SQL (Structured Query Language) to write a query that will return to us the tiny fraction that we need to use for this lab. First, let us break down what a generic SQL SELECT query looks like:

```
SELECT feature_id, site_index, species_cd_1, species_pct_1, geom
FROM vancouver_island_vri
WHERE bec_zone_code = 'CWH';
```

The **SELECT** keyword is the most common database operation and the only one that we will use for this lab, but understand that there are keywords for other tasks in the database like creating tables, inserting data, updating data, and deleting data, to name a few. By convention, SQL keywords are capitalized, but most software including ArcGIS Pro do not require this.

What follows the **SELECT** keyword is a comma-separated list of column names that we want to retrieve from the table, `feature_id`, `site_index`, `species_cd_1`, `species_pct_1`, `geom`. You could also replace these with `*` to return all of the column names and this is generally the default behavior across ArcGIS Pro, so for most tasks in ArcGIS Pro you generally will not need to specify any column names when writing a SQL query.

- `feature_id` is the primary key of the VRI table, this uniquely identifies all of the polygons from each other. When we select any subset of a table, we always want to grab the primary key because this allows us to keep track of which polygon we are dealing with from the originating table.
- `site_index` is the estimate of site productivity for tree growth for the polygon; the values are recorded as decimal numbers.
- `species_cd_1` is the leading tree species code for the polygon; the values are recorded as text strings.
- `species_pct_1` is the percentage of the polygon occupied by the leading species represented by the `species_cd_1`; the values are recorded as integers. Note that “occupy” here is defined differently, depending on whether the stand is young or mature.
- `geom` is the geometry column, which is where the coordinates of the polygons are actually stored. We need this column if we want to map the results of our query.

There are 190 other attributes (columns) in the VRI table for each polygon,

so this is just a small subset. You should review the VRI Relational Data Dictionary if you have any doubts about what a particular field refers to.

Next, the statement describes the table name in the database that these columns are being selected from, `FROM veg_comp_lyr_r1_poly_2025`.

The `WHERE bec_zone_code = 'CWH'` clause is a conditional statement, which limits the rows (features) returned from the originating table. Note here that we can refer to any column names from the originating table, even if they do not appear in the list of column names that we want to return, as long as they exist in the originating table. Finally, all SQL queries are always concluded with a semicolon `;`.

You should think of `SELECT` as *subsetting columns of a table* (along the x-axis) and `WHERE` as *subsetting rows of a table* (along the y-axis). When used together, we are reducing both dimensions of the originating table simultaneously, which substantially reduces the total data that we are retrieving. We will use this principle and expand the SQL statement above in the next step in order to retrieve only the tiny fraction of the VRI data that we need.

Step 4: From the top ribbon in ArcGIS Pro, navigate to the “Map” tab, then click the down arrow under “Add Data” and select “Query Layer...” from the drop-down menu. In the “New Query Layer” dialogue window that appears, click the “Datasource” drop-down menu at the top and you should see the UBC PostgreSQL server connection to the `vri` database listed there that you added at the beginning of this task. It should look something like “PostgreSQL-FRST-GIS-vri(student).sde”.

If not, then click the little database connection button next to the drop-down menu in the dialogue window to add the database connection. (Note: in this specific case, you will need to add the port number to the path string like `FRST-GIS.ead.ubc.ca,5432`)

We are going to write a select query to retrieve these data from the server, but we do not need all the columns, nor all the rows. We only need polygons that are managed forests and for those polygons, we only need a few attributes that we are going to use later in the lab (“`bec_zone_code`”, “`bec_subzone`”, and “`proj_age_1`”).

Step 5: Toggle on “List of Tables” to see the table names and then click on the VRI table in the list “`vri.public.vancouver_island_vri`” to show the list of columns. (If you get an error message here, then save your project, restart ArcGIS Pro and try again). We are going to write a SQL query to select the polygons that meet some of our initial criteria and then return the values of the columns (attributes) that we will need later. In the “Name” field, name the new query layer “`vancouver_island_vri_select`”.

The VRI data include descriptions about both forested and non-forested lands in British Columbia. Since we are only concerned with forests, we are going to only find polygons that are managed forests. We can do this simply with

`SELECT * FROM vancouver_island_vri WHERE for_mgmt_land_base_ind = 'Y'`. See here that we have not defined any specific column names to return, `SELECT *` means *give me all the columns*, so in this example, we are only subsetting on the rows with `WHERE for_mgmt_land_base_ind = 'Y'`, but not the columns. If instead we used `SELECT for_mgmt_land_base_ind FROM vancouver_island_vri`, then we would get all the polygon values for only the `for_mgmt_land_base_ind` column name (subset columns, but not rows). Run the query and inspect the output as a new layer.

Q1. What does `for_mgmt_land_base_ind` select for? What parameters are required to receive a “Yes” value? Hint: take a look at the VRI Relational Data Dictionary. (10 points) Pro tip: When a column value stores a text string, we use single quotation marks in SQL `'abc'` around the values that we are searching for; double quotation marks `"abc"` will fail for a SQL query.

Step 6: Using what you just learned, draft a select statement that limits the VRI polygons to only managed forests and return only the column names `feature_id`, `bec_zone_code`, `bec_subzone`, `proj_age_1`, and `geom`. Press the “Validate” button on the bottom of the dialogue window. If the statement validates successfully, then the “Next” button will allow you to proceed.

Pro tip: If you are working from a laptop, be sure to plug into your power source as your computer power options may significantly slow down your ArcGIS Pro performance to conserve battery.

Step 7: In the next dialogue, toggle `feature_id` as the field that uniquely identifies all features in the table, “Unique Identifier Field(s)” (i.e., the primary key), leave the other fields toggled off. Once you click “Finish”, your select statement will be applied and sent to the server. You will see a small dialogue window that indicates that the extent is being calculated, and this should take about 3-5 minutes before you will see a layer appear in your Table of Contents. Read on to understand the next steps while the server processes your query.

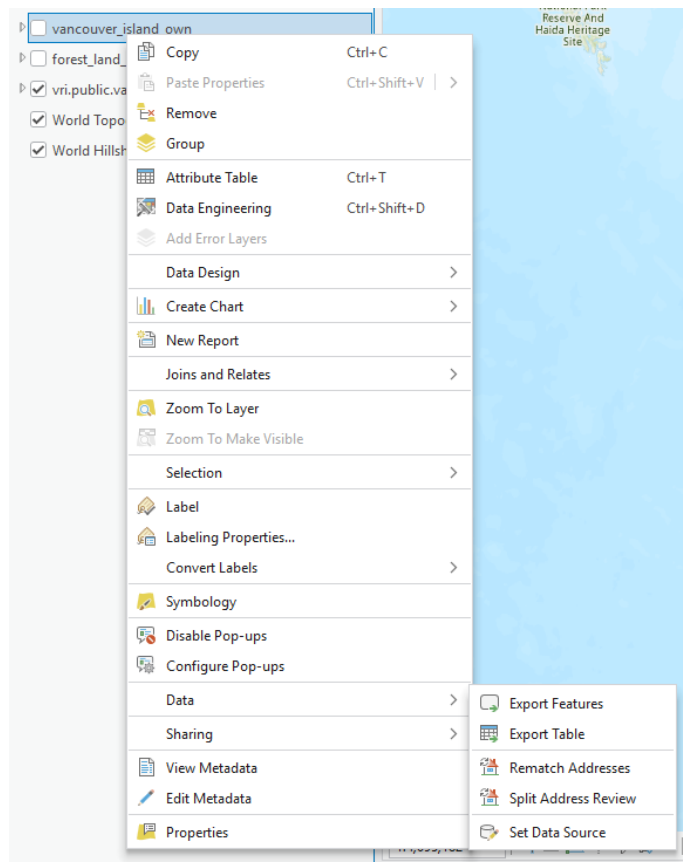
Pro tip: It will take another 3-5 minutes before you will see any polygons drawn on your map. You can speed some steps up if you “Pause” the map drawing mode by pressing the pause button at the very bottom of your map. In general, try to keep large layers toggled off so that they are not drawing in the background.

Save your ArcGIS Pro project now.

Our goal here is not necessarily to visualize the results, though you should see you map automatically pan to Vancouver Island. You can toggle the layer off and visualize it later. If you open the attribute table, you should see the five columns we requested and there should be 250,000+ polygons. (Click the “Load All” button at the bottom of the attribute table to verify, which will take some additional time).

The query layer that you just created is persistent between ArcGIS Pro sessions if you saved and re-open the project later, but the query layer does not actually contain any data because it just provides the instructions to apply the query to the remote database. It will re-run the query each time you re-open your ArcGIS Pro project, which will take some time to retrieve the data. To avoid this, we are going to actually download the data to our local computer.

Step 8: Export the selection you have made your ArcGIS Pro project geodatabase by right-clicking the “vancouver_island_vri_select” layer in your Table of Contents, select “Data”, then select “Export Features”. In the dialogue window that opens, change the output filename to “forest_land_base” (Note: layers stored in geodatabases do not have file extensions), the default location is already the geodatabase for your ArcGIS Pro project. Click “Ok”. This should take about 5 minutes to complete.



Next, we are going to export the other layers and download them directly to our ArcGIS Pro project geodatabase.

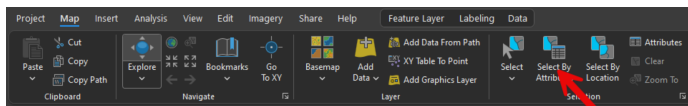
Step 9: From the “Catalog Pane”, export the “vancouver_island_own”

layer to your ArcGIS Pro project geodatabase by right-clicking the “vancouver_island_own” layer, select “Export”, then select “Export Features...”. In the dialogue window that opens, change the output filename to “vancouver_island_own” (Note: layers stored in geodatabases do not have file extensions), the default location is already the geodatabase for your ArcGIS Pro project. Click “Ok”. This should take a few minutes to complete.

Repeat the last step for “vancouver_island_human_disturbance” and “vancouver_island_landscape_units”. Altogether, this is about 600 MB of data, so be sure you have sufficient disk space and be patient as the data transfer completes. When finished, you can drag the layers into ArcGIS Pro to visualize them on the map. If you paused drawing mode earlier, then unpause it now.

The cumulative effects framework only considers forests located on crown land. Next, we will use the forest ownership layer “vancouver_island_own” to select for land ownership types. The `ownership_description` field describes whether the polygon is private or crown land in a text format. There are lots of different types of crown ownership categories, but for our purpose we need all of them. To do this quickly, we can use `LIKE`, which is a pattern-matching operator for strings. We need to use the `%` wildcard to partially-match some pattern. For example, we can search for any row in “vancouver_island_own” with a description of “Crown” using the following statement: `SELECT * FROM vancouver_island_own WHERE ownership_description LIKE 'Crown%'`. Basically `%` is a wildcard for zero or more characters, so when we construct the pattern `'Crown%'`, we are saying *match any string that starts with “Crown” and has any number of characters after it*.

Step 10: Open the attribute table of “vancouver_island_own” by right-clicking the layer in the contents pane. Then click the “Select by Attributes” button then toggle the SQL Editor button to show the SQL codeblock. Write a query to select all the polygons that are crown land as described in the previous step. Note that in this particular interface, you only need to give the conditions for selection, so you can omit `SELECT * FROM table WHERE`. Export the features to your ArcGIS Pro project geodatabase as you did with the the vancouver_island_vri_select, naming the new layer “crown_land” and ensuring that “Use the selected records” is toggled on.



You can read more about the CEF human disturbance layer and methodology here: <https://catalogue.data.gov.bc.ca/dataset/bc-cumulative-effects-framework-human-disturbance-current>

Step 11: Next, we will select the areas where crown land overlaps with potentially forested areas.

Use the **Pairwise Intersect** tool. You can search for it by clicking the toolbox

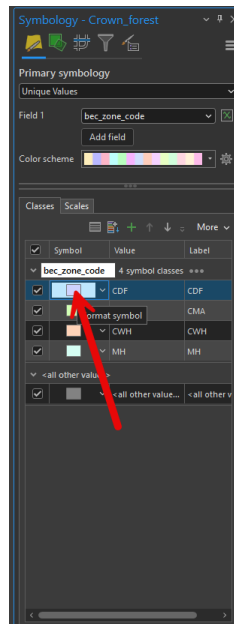
on the top ribbon (open the “Analysis” tab, then click “Tools”).

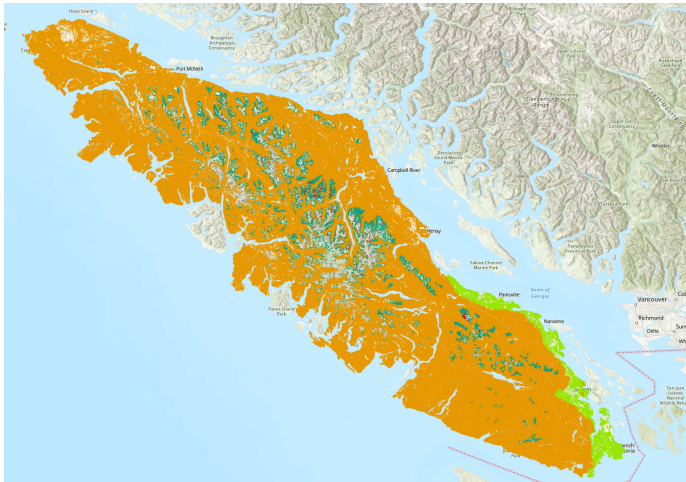
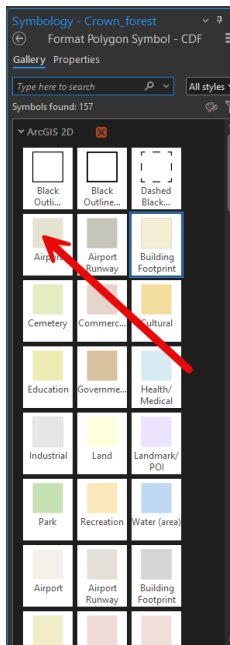
- Input: forest_land_base, crown_land
- Output: Crown_Forest

Once the tool has finished running, open the new Crown_Forest attribute table to ensure that “bec_zone_code” and “bec_subzone” attributes have been preserved.

Step 12: Turn off visibility for other layers in the contents pane on the left side. If the contents pane has been closed, it can be reopened from the ribbon at the top of the screen. Visualize the Crown_Forest layer by `bec_zone_code` or `bec_subzone`. To do this, right-click the layer in the contents pane and select “Symbology”. Change the primary symbology to “Unique Values” and “field 1” to the attribute of your choice.

Pro tip: Polygon outlines can be removed by clicking the symbol of each row in the “Classes” table of the symbology layer. When given the option, select a symbol that does not have an outline. This makes it easier to see the area of many smaller polygons combined.





Step 13: In BC forestry standards, the age of a forest stand is categorized by “Seral Stage.” Since the required ages for mature or old growth forests varies depending on the BEC zone, seral stages are used to denote the relative age class of a stand; “Early”, “Mid”, “Mature”, and “Old”. We will create a new field and classify each polygon depending on its estimated age.

Open the **Crown_Forest** attribute table and select “Add” in the table ribbon. This will take you to the field view. At the bottom, create a new field using:

- Name: “Seral_Stage”

- Data Type: Text

To save this new field, click “Save” on the Ribbon at the top of the window. Once it is saved you can close the field view and return to the **Crown_Forest** attribute table. Once it is saved you can close the field view and return to the **Crown_Forest** attribute table.

Step 14: Right-click the new field and select “Calculate Field”. The age categories for each seral stage depend on which BEC zone the stand falls within. These age categories correspond with the growth rate and productivity of different eco-zones. The provided code defines the breakdown of Early, Mid, Mature and Old for the three BEC zones found on Vancouver Island.

In the expression field paste the following code. This takes the BEC zone code and estimated age as inputs into a function that we will define.

In the code block under “Seral_stage =”, paste:

```
getSeralStage(!BEC_ZONE_CODE!, !PROJ_AGE_1!)
```

In the “Code Block” paste:

```
def getSeralStage(bec, age):

    if bec == 'CWH':
        if age <= 40:
            return 'Early'
        elif age <= 80:
            return 'Mid'
        elif age <= 250:
            return 'Mature'
        else: return 'Old'

    elif bec == 'MH':
        if age <= 40:
            return 'Early'
        elif age <= 120:
            return 'Mid'
        elif age <= 250:
            return 'Mature'
        else: return 'Old'

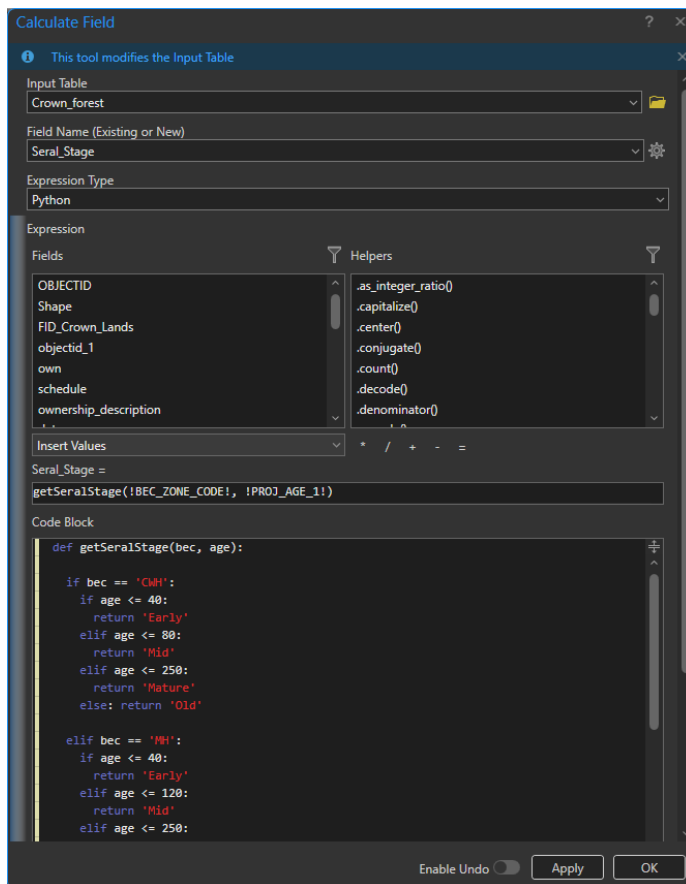
    elif bec == 'CDF':
        if age <= 40:
            return 'Early'
        elif age <= 80:
            return 'Mid'
        elif age <= 250:
```

```

        return 'Mature'
    else: return 'Old'

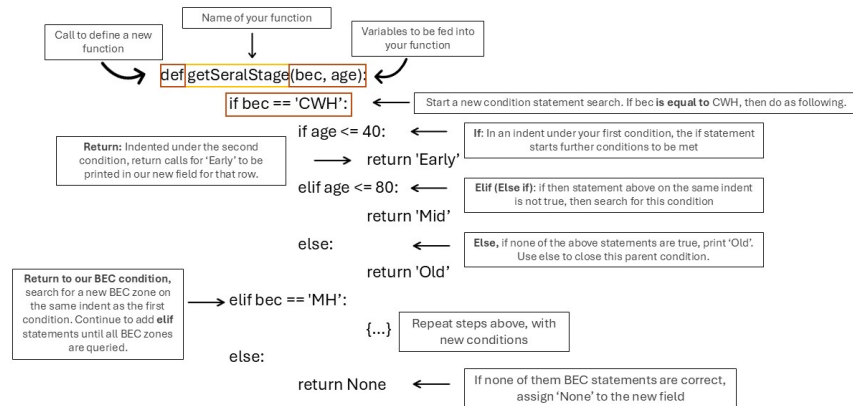
else:
    return None

```



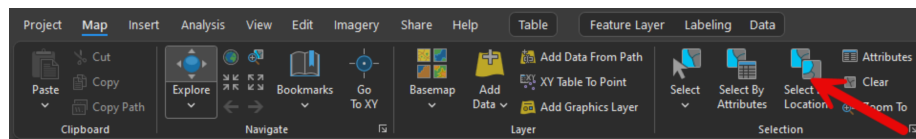
Note: Ignore the warning message that appears.

To better understand what is being done in this code chunk, the syntax is described below.



Check that all the rows have been populated in the `Seral_stage` field.

Q2. Define “old growth” and cite at least one primary source. What characteristics might be overlooked by a definition of “old growth” that only accounts for stand age? (10 points) **Step 15:** Our current layers only indicate where crown forests *could potentially occur*. In this step, we will reset the seral stage in areas that have experienced human disturbances such as roads, buildings, or logging. We are not removing these areas entirely, as they remain part of the landscape unit where old-growth forest could exist if it had not been cleared. We will use `vancouver_island_human_disturbance` to do so. Select by attribute where `cef_human_disturb_flag` is equal to `Human Disturb Current 20yr`. Export the selected attributes as a new feature named `Disturbed_areas`.



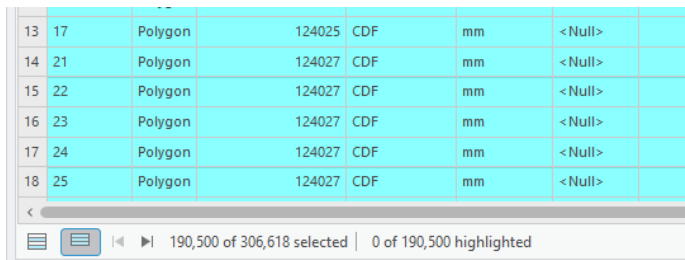
Then, use **Select By Location** in the top ribbon to identify areas of overlap between potential crown forest stands and areas of disturbance.

- **Input Features:** `Crown_Forest`
- **Relationship:** `Intersect`
- **Selecting Features:** `Disturbed_areas`
- **Selection Type:** `New Selection`

Leave other options with their defaults. This may take a couple of minutes to complete.

Step 16: In the Crown_Forest attribute table:

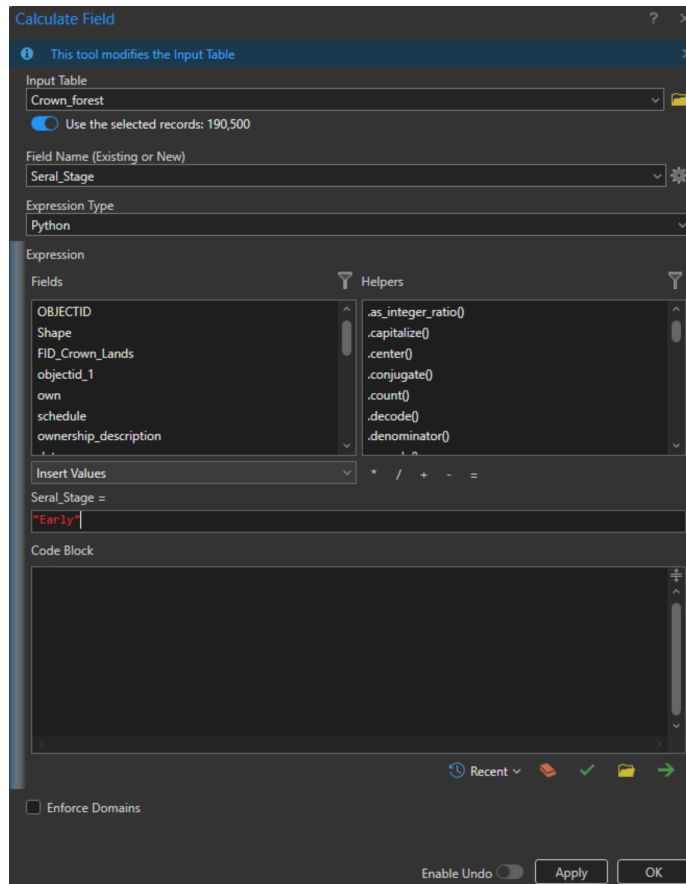
- Click “Show Selected Records”



13	17	Polygon	124025	CDF	mm	<Null>	
14	21	Polygon	124027	CDF	mm	<Null>	
15	22	Polygon	124027	CDF	mm	<Null>	
16	23	Polygon	124027	CDF	mm	<Null>	
17	24	Polygon	124027	CDF	mm	<Null>	
18	25	Polygon	124027	CDF	mm	<Null>	

190,500 of 306,618 selected | 0 of 190,500 highlighted

- Right-click the Seral_Stage field, then click “Calculate Field”
- **Expression:** "Early"
- Ensure that **Use the selected records** is toggled on
- Remove your previous code text if it still remains. Type “Early” (with quotes) into the expression box and leave the Code Block empty.
- Click “Apply”
- Click “Clear” to remove selection



Pro tip: Always clear your selection once you are done working with it. This prevents potential missteps in later steps.

Step 17: Old growth forest quantities are assessed in terms of Land Units and BEC Subzones. Land Units are land parcels pre-determined by the BC Government for ease of land management and assessment. We will need to divide our current Crown Forest Landbase by the pre-determined Landscape Units.

Open the **Pairwise Intersect** tool again:

- Input: Crown_Forest, vancouver_island_landscape_units
- Output Name: CrownForest_LU

Task 2: Calculate Percentage of Old Growth Forests

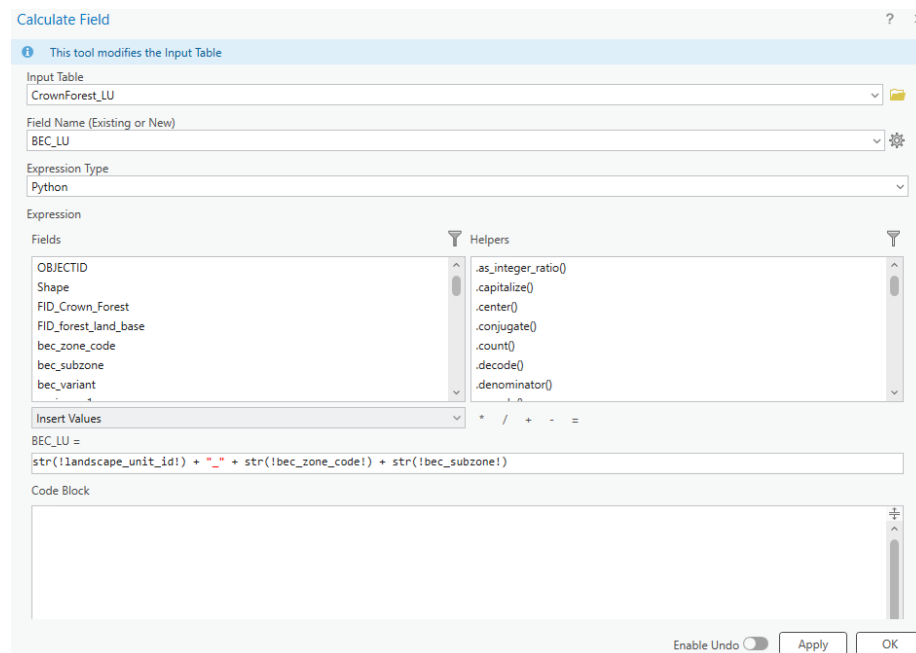
Step 1: Now that we have prepared our layers, we will calculate the percentage of old growth forest within each land unit. To do this we will create a key that uniquely identifies each Landscape Unit + BEC zone + BEC subzone combination. This is a good opportunity to close any unneeded attribute tables or turn off layer visibility.

Add a new field to CrownForest_LU attribute table:

- Add a new field to **CrownForest_LU**
- Name: **LU_BEC**
- Type: **Text**

In the **LU_BEC =** textbox, paste:

```
str(!landscape_unit_id!) + "_" + str(!bec_zone_code!) + str(!bec_subzone!)
```

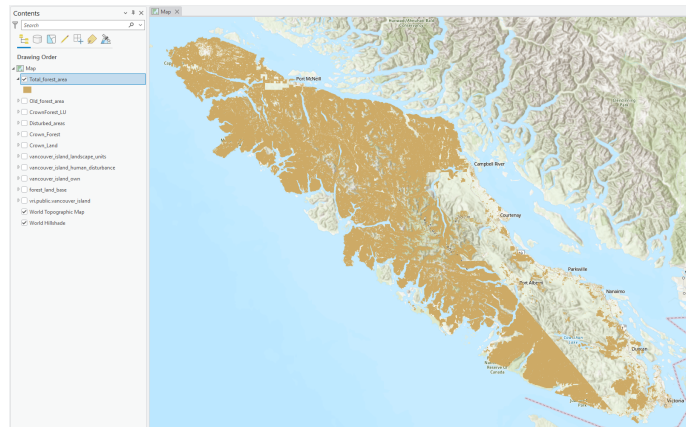


The **str()** function ensures that the attribute values are treated as text strings rather than numeric values for addition. The result should be a new string of landscape unit, BEC zone and subzone.

Step 2: We will now calculate the total amount of forested area in each of these landscape unit/BEC zone combination. With the **Dissolve** tool, we can create new polygons from the landscape identifier:

- Input Table: **CrownForest_LU**
- Dissolve Fields: **LU_BEC**
- Statistics Fields: **Shape_Area** as SUM and **bec_subzone** as FIRST
- Output Table: **Total_Forest_Area**
- Check “Create Multipart Features”

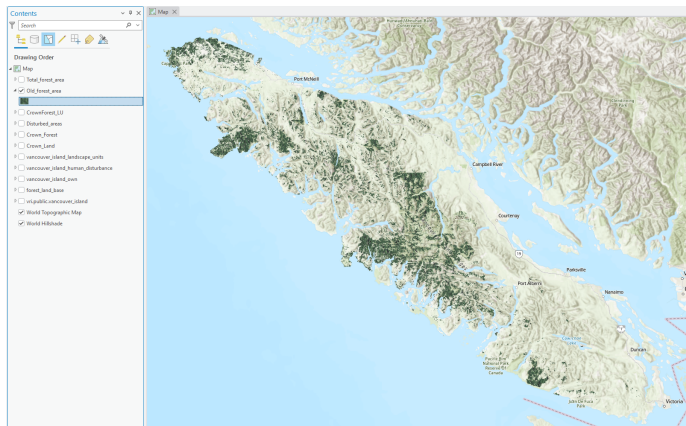
This creates a polygon layer where each feature is one LU_BEC area with total crown forest area as an attribute.



Step 3: To identify areas of assumed old growth, we will calculate the area of only forests where **Seral_stage** equals **old**. We can select for old forest by using **Select by Attributes on CrownForest_LU** before using the **Dissolve** tool. Ensure that the “Use the selected records” option is switched on.

- Select by Attribute: **Seral_Stage = "Old"**
- Dissolve Fields: **LU_BEC**
- Statistics Fields: **Shape_Area** as SUM
- Output Table: **Old_Forest_Area**
- Ensure **Use the selected records** is on
- Check “Create Multipart Features”

Now we have created a table of total crown forest area and a table of old growth crown forest area, each by landscape unit.



Step 4: We can join these tables back together using a

Use **Join Field** with the following expression:

- Input Table: Total_Forest_Area
- Input Field: LU_BEC
- Join Table: Old_Forest_Area
- Join Field: LU_BEC
- Transfer Field: Shape_Area (from Old_Forest_Area)

Geoprocessing

Join Field

This tool modifies the Input Table

Parameters Environments

Input Table
Total_forest_area

Input Field
LU_BEC

Join Table
Old_forest_area

Join Field
LU_BEC

Transfer Method
Select transfer fields

Transfer Fields
Shape_Area

Index Join Fields
Do not add indexes

Validate Join

We now have two fields in `Total_forest_area` named `Shape_Area` which could lead to some confusion. We can assume that the column with the smaller numbers is the old growth forest area. This should be the last column in the attribute table. Right-click the attribute name to open the field view. Rename the alias of `Shape_Area` to `Total_Area` and the alias of `Shape_Area_1` to `Old_Area`. Save your changes.

Step 5: Calculate Percent Old Growth

Add a new field to `Total_Forest_Area`:

- **Name:** PercentOld
- **Type:** Double

Use **Calculate Field** with:

- **Expression:**

```
calcPercent(!Shape_Area_1!, !Shape_Area!)
```

Note: the expression still uses the column name rather than the alias.

- **Code Block:**

```
def calcPercent(old, total):
    if old is None or total is None or total == 0:
        return 0
    return (old / total) * 100
```

Step 6: Classify Percent Compared to Provincial Targets.

The target percentages of old growth forests are dependent on the BEC subzone. Instead of assigning each LU-BEC to a discrete class (Low / Intermediate / High), in this step you will create a numeric field representing the difference between the calculated percent old growth and the subzone-specific “high” threshold. You will do this by writing your own python expression to calculate the difference. Refer to the code in step 10 for a guideline on how to build the function. This difference (PercentOld - HighThreshold) is what you will visualize as a continuous gradient:

- Positive values exceed the high target,
- Negative values fall short of the high target,
- Zero meets the high target exactly.

Begin by adding a new field to **Total_Forest_Area**:

- Name: DiffFromHigh
- Type: Double

The percentage thresholds for each class is as follows:

Subzone Code	Name	Low (less than):	High (greater than):
CDFmm	Coastal Douglas-fir	9	13
CWHmm	Coastal Western Hemlock Moist Maritime	9	13
CWHvh	Coastal Western Hemlock Very Wet Hypermaritime	13	19
CWHvm	Coastal Western Hemlock Wet Hypermaritime	13	19
CWHxm	Coastal Western Hemlock Very dry maritime	9	13
MHmm	Mountain Hemlock Moist Maritime	19	28

Use **Calculate Field** with:

- Expression:

```
diff_from_high(!LU_BEC!, !PercentOld!)
```

Now write your own expression to calculate the difference between the old growth target and the calculate old growth percentage.

Screenshot 1. Upload a screenshot of your python expression (10 points) **Pro Tip:** The subzone code can be isolated by removing the first 5 characters with a slice. This would look something like this:

```
bec = lu_bec[5:]
```

We read this as *give me all the elements from position 5 to the end of the string*, where the colon `:` separates the “from” and “to” values. If we leave one side of the colon blank, then we are saying we want whatever the minimum (left side of colon) or maximum (right side of colon) number of elements is. Python indexing starts at zero, so if I want to return the letter “g” from the alphabet string `abcdefghijklmnopqrstuvwxyz`, then I would write something like `alphabet[6]` or if I want all letters before “g”, then `alphabet[:5]`.

OBJECTID	Shape	LU_BEC	Shape_Length	Total_Area	Old_Area	PercentOld	DiffFromHigh
1	Polygon	1627_CWHvh	685934.187898	404157323.892434	140984601.454016	34.883594	15.883594
2	Polygon	1627_CWHvm	664612.509813	414153045.770105	15580246.981542	3.761954	-15.238046
3	Polygon	1627_MHvm	12089.612881	594224.522068	<Null>	0	-28
4	Polygon	1630_CWHvh	1632934.364127	268270282.514044	139324178.128542	51.934257	32.934257
5	Polygon	1630_CWHvm	44873.699209	21992383.857746	2302402.51689	10.46909	-8.53091
6	Polygon	1631_CWHvh	530838.309977	136569462.899258	84311044.888369	61.734917	42.734917
7	Polygon	1638_CWHvh	274864.403272	79633222.740218	38746294.575689	48.655942	29.655942
8	Polygon	1638_CWHvm	238860.100607	127104597.351475	31247096.542911	24.583766	5.583766
9	Polygon	1640_CWHvm	908678.910536	324949651.421247	12663930.244541	3.897198	-15.102802
10	Polygon	1640_MHvm	159942.014153	16342406.595165	2676229.500521	16.375982	-11.624018
11	Polygon	1641_CWHvh	179755.988322	39448092.917205	9683010.930447	24.546208	5.546208
12	Polygon	1641_CWHvm	704716.178475	322452457.354111	12578936.639955	3.901021	-15.098979

Step 7: Visualize Results

In the **Contents** pane, right-click the **Total_Forest_Area** layer and select **Symbology**. In the Symbology panel, set the **Field** to **DiffFromHigh**. This will display the data based on the values in that field. Under **Primary symbology**, choose **Graduated Colors** (a gradient).

Select a **color ramp** that makes sense for your data. For values that represent *low to high* differences, use a single-hue light-to-dark ramp (e.g., light yellow to dark green). Avoid using colors that are difficult to interpret (e.g., bright rainbow colors). - Think about your audience—pick colors that clearly show the trend and are readable to someone with color-vision deficiencies. Adjust the **classification method and number of classes** if needed (e.g., Natural Breaks, Quantile). This determines how your data values are grouped into color ranges.

Step 8: Create the Layout Map

From the top ribbon, go to the **Insert** tab and select **New Layout**. Choose an appropriate page size (e.g., Letter 8.5x11 or A4). Use the **Map Frame** tool to insert your map view into the layout. Resize and position it so it fills most of the page. - Add essential map elements: - **Title**: Insert a descriptive map title (e.g., “*Difference from High Threshold in Total Forest Area*”).

- **North Arrow**: select “Insert”, then select “North Arrow”. Place it in a

corner where it does not overlap important data.

- **Legend:** select “Insert”, then select “Legend”. This should explain the symbology of your `DiffFromHigh` values.
- **Scale Bar:** select “Insert”, then select “Scale Bar”. A useful addition if map distance matters.
- **Author/Date/Text:** Add your name and date for documentation.

Rearrange and resize elements so the layout looks clean, balanced, and easy to read. Avoid cluttering the map with overlapping items.

Step 9: Create the Layout Map

When you are satisfied with the layout, go to **select “Share”, then select “Export Layout”**. Choose **PNG** as the file format.

Q3. Describe the results of your final output. What is the distribution of old growth targets met across Vancouver Island? (10 points)

Q4. Select 3 other fields in the VRI layer and briefly discuss how they could provide further insight into current old growth status. (10 points)

Q5. How is VRI data collected and what limitations does it pose? (10 points)

Q6. How might the scale and resolution of GIS data affect the accuracy of old growth assessments? (10 points)

Table 1. A table with fields for total area, old-growth area, and percentage old growth. (10 points)

Map 1. A map showing the study area and the distribution of difference in calculated old-growth percentage and targets, by polygon. (20 points)

References

British Columbia & BC Environment (Eds.). (1995). *Biodiversity guidebook*. Forest Service, British Columbia: BC Environment.

Canadian Institute of Forestry (2022). *Old Growth Information bulletin*. March, 2022

CBC. (2022, February 11). Fairy Creek protesters bid to have charges stayed. *CBC News*. <https://www.cbc.ca/news/canada/british-columbia/protest-court-proceedings-1.6348014>

Forest Analysis and Inventory Branch (2025). VRI - 2025 - Forest Vegetation Composite Rank 1 Layer (R1). British Columbia Data Catalogue.

LePage, P., & Banner, A. (2014). Long-term recovery of forest structure and composition after harvesting in the coastal temperate rainforests of northern British Columbia. *Forest Ecology and Management*, 318, 250–260. doi.org/10.1016/j.foreco.2014.01.031

Ministry of Water, Land and Resource Stewardship (WLRS). (2024). Old Growth Forests in British Columbia: Cumulative Effects Assessment Backgrounder. Victoria, British Columbia.

Ministry of Water, Land and Resource Stewardship. 2024. Old Growth Forest Management in British Columbia: Provincial Backgrounder. Victoria, British Columbia.

Summary

In this lab you standardized VRI layers, filtered the Forest Management Landbase, calculated old-growth areas, and derived percentages to compare across polygons. You then produced a map and exportable table to communicate your findings. The key skills reinforced were: consistent data preparation, careful attribute logic for inventory fields, transparent calculation workflows, and interpreting results within an ecological and management context.

Return to the **Deliverables** section to check off everything you need to submit for credit in the course management system.

Chapter 4

Remote Sensing Imagery Analysis

Written by Paul D. Pickell

Lab Overview

The aim of this lab is to learn about the electromagnetic spectrum (EMS), understand spectral properties of different surfaces, and get comfortable using ArcGIS Pro to load and explore different types of remotely sensed images, display individual spectral bands, make different colour composites, and view spectral signatures. In addition, you will calculate two difference spectral indices representing built up areas and green vegetation.

Learning Objectives

- Differentiate regions of the electromagnetic spectrum for urban applications
 - Use Landsat 9 spectral reflectance to map vegetation and urban areas
 - Composite bands together to create true colour and false colour visualizations
 - Analyze spectral signatures of different land cover types
-

Deliverables

Answers to the questions posed throughout the lab. (31 points)

Screenshot of your spectral plot. (2 points)

Screenshot of you band arithmetic equation for NDVI. (2 points)

Four screenshots of your false-colour composites. (40 points)

Map that shows your High_Vegetation_Areas and Urban_area layers overtop the provided Landsat Imagery. Pick appropriate colours and transparency for each layer. (25 points)

Data

We will be working with a multispectral image of Vancouver from the Landsat 9 satellite that is available to download from the course management system or you download it yourself from EarthExplorer using the scene id: LC09_L2SP_047026_20260807_20260808_02_T1. This is a Collection 2, Level 2 Science Product, which means that the provider has already processed the raw satellite imagery and corrected for atmospheric effects.

Task 1: The EMS

The electromagnetic spectrum (EMS) is the distribution of electromagnetic radiation according to wavelength/frequency, and includes radio waves, visible and infrared light, x-rays, gamma rays, and more. In remote sensing, we use the reflective, absorptive, and emissive properties of terrestrial features to identify and measure them (i.e., how do different wavelengths in the EMS interact with the surface of the Earth?).

Note: It is important to recognize that the visible part of the EMS is the only section that humans can see. All colours in the visible spectrum are wavelengths, but not all wavelengths in the EMS are colours.

Spectral Reflectance

Figure 1 shows the reflective characteristics of various features of the earth's surface. Use this figure to answer **Q1 – Q4**.

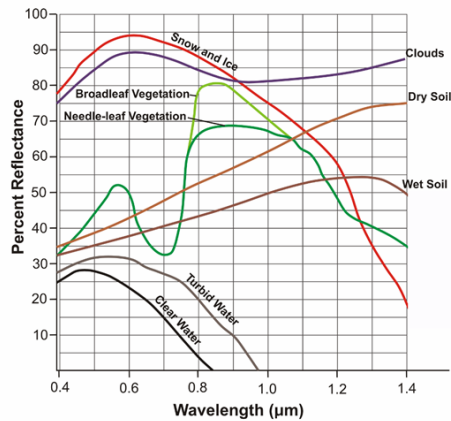


Figure 1: Reflectance characteristics of various features at different wavelengths.

Q1. For broadleaf and needle-leaf vegetation, what is the approximate wavelength that is reflected most, and what section of the ems does this range belong to? (1 point)

Q2. Soil and vegetation reflect roughly the same proportion of blue light. (1 point)

Q3. Give a wavelength (in microns) at which snow and ice, dry soil, and vegetation are indistinguishable by their reflectance. In other words, at which wavelength is the proportion of radiation reflected the same (+/- 10%) for these features? (1 point)

Q4. Broadleaf and needle leaf vegetation reflect the same amount at 0.7 microns. What causes this? Is there something contained in the foliage of both types of vegetation which causes identical spectral signatures? How does this pattern in spectral reflectance affect how we see live vegetation? (4 points)

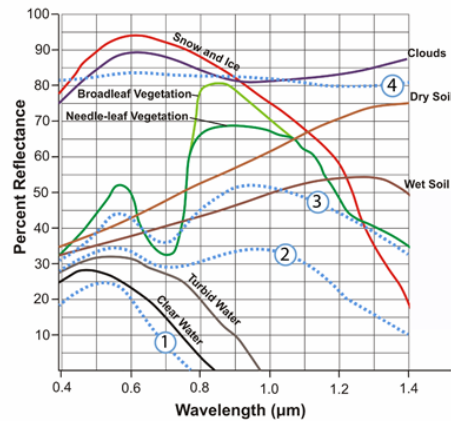


Figure 2: Reflectance characteristics of unknown features.

Q5. Figure 2 contains 4 additional spectra, belonging to unknown surface features. Hypothesize about what these spectra might be and provide your reasoning. Use the known features (broadleaf vegetation, wet soil, etc.) and what you have learned from class/readings to inform your choices. This is a difficult task, and educated guesses are all that is asked for. Do a bit of research, put some thought into it, and explain the reasoning for your guesses. These spectra do not represent the features which are already labeled. You must think of new features which could be observed with remote sensing. (4 points)

Task 2: Landsat 9 Bands, the EMS & ArcGIS Pro Software

Table 2: Parameters of Landsat 9's Operational Land Imager (OLI) sensor

Band	Wavelength Range (microns)	Spectral Region	Spatial Resolution (meters)	
1	0.43-0.45	Coastal/Aerosol	30	Coastal and aerosol
2	0.45-0.51	Blue	30	Coastal water
3	0.53-0.59	Green	30	
4	0.64-0.67	Red	30	Chlorophyll
5	0.85-0.88	Near Infrared	30	Biomass
6	1.57-1.65	Short-wave Infrared 1	30	Vegetation and soil
7	2.11-2.29	Short-wave Infrared 2	30	
8	0.50-0.68	Panchromatic	15	Sharper image
9	1.36-1.38	Cirrus	30	
10	10.60-11.19	Thermal Infrared 1	100	Thermal mapping
11	11.50-12.51	Thermal Infrared 2	100	Improved thermal mapping

Q6. Each pixel of the thermal infrared bands 10-11 covers _____ pixels of the other bands. If it helps, draw a picture of the two pixel resolutions. (1 point)

Q7. Bands 10-11 are recorded with a coarser resolution because thermal radiation has a very _____ wavelength. Therefore, there is _____ energy available to sense. (1 point) **Step 1:** Start a new ArcGIS Pro Project. Download the zip data from the course management system and unzip the LC09_L2SP_047026_20260807_20260808_02_T1.zip file into your new ArcGIS Pro project folder. This location is usually C:/Users/yourusername/Documents/ArcGIS/Projects. Look at the contents that have been extracted. These are all the files that you would receive if you downloaded the image yourself, but we will not use all of them for this lab. Importantly, you should notice that the files follow a naming convention. Use the tables below to reference the correct files in later steps. Note that Band 8 (Panchromatic), Band 9 (Cirrus Cloud), and Band 11 (TIRS 2) are not processed for surface reflectance, so you will not see them here.

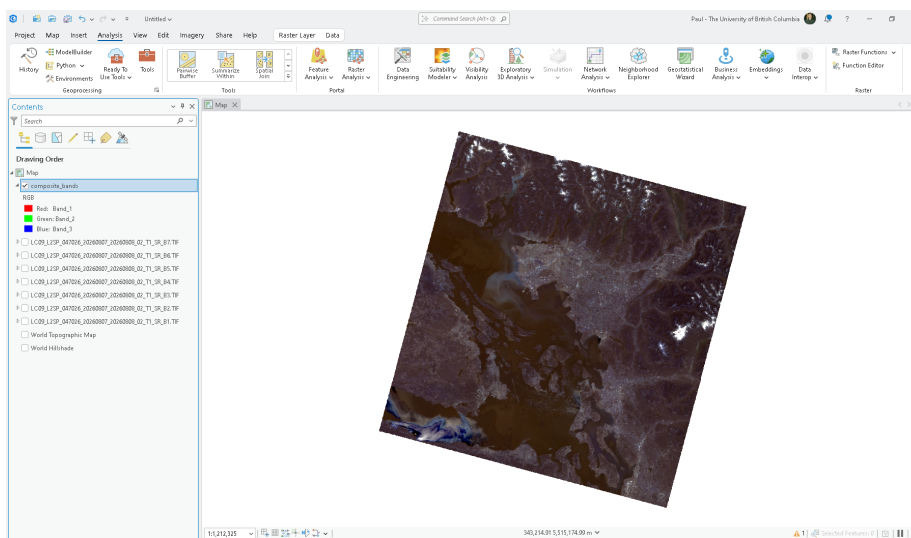
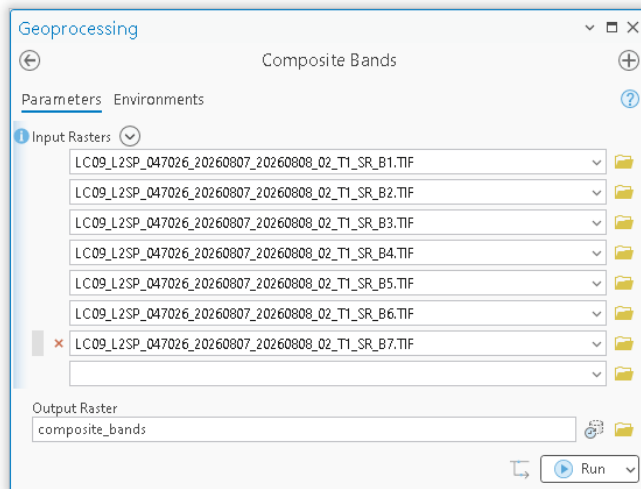
File Name Element	Value	Interpretation
Sensor	LC09	Landsat 9, OLI-2/TIRS-2 sensor
Processing level	L2SP	Level-2 Science Product (Surface Reflectance and Surface Temperature)
WRS Path/Row	047026	Path 047, Row 026 (defines the scene's ground location)
Acquisition date	20260807	Date the image was captured (August 7, 2026)
Processing date	20260808	Date USGS generated this product (August 8, 2026)
Collection	02	Collection 2 processing
Tier	T1	Tier 1 - highest quality, precision terrain-corrected

File Suffix	Description
MTL.txt / MTL.xml	Metadata file - sensor settings, corner coordinates, radiometric/reflectance
ANG.txt	Angle coefficients file - used to derive per-pixel solar/sensor
QA_PIXEL.TIF	Pixel quality bitmask - quality assurance (QA) flags for cloud, cloud
QA_RADSAT.TIF	Radiometric saturation QA - flags saturated pixels
SR_B1.TIF	Surface Reflectance, Band 1 (Coastal/Aerosol)
SR_B2.TIF	Surface Reflectance, Band 2 (Visible Blue)
SR_B3.TIF	Surface Reflectance, Band 3 (Visible Green)
SR_B4.TIF	Surface Reflectance, Band 4 (Visible Red)
SR_B5.TIF	Surface Reflectance, Band 5 (Near-Infrared)
SR_B6.TIF	Surface Reflectance, Band 6 (SWIR 1)
SR_B7.TIF	Surface Reflectance, Band 7 (SWIR 2)
SR_QA_AEROSOL.TIF	Aerosol QA - flags used in the atmospheric correction for
ST_B10.TIF	Surface Temperature, Band 10 (Thermal, in Kelvin)
ST_ATRAN.TIF	Atmospheric transmittance - used in the surface temperature
ST_CDIST.TIF	Pixel distance to nearest cloud
ST_DRAD.TIF	Downwelling radiance
ST_URAD.TIF	Upwelling radiance
ST_TRAD.TIF	Thermal radiance (top-of-atmosphere, before surface temperature
ST_EMIS.TIF	Emissivity, from ASTER Global Emissivity Database
ST_EMISD.TIF	Emissivity standard deviation
ST_QA.TIF	Surface Temperature uncertainty (in Kelvin)

Step 2: Add the geotiffs for Bands 1-7 to your map. Toggle them on/off, pan around, and zoom in to inspect different locations. Notice that each band is a grey scale image, even the images that represent the visible part of the spectrum. Notice how a land cover might appear bright in one band and dark in another. Pixels that are bright have high **reflection** of light back to the sensor in this band of the spectrum. Pixels that are dark have high **absorption** in this band of the spectrum.

We are going to composite these bands together into a new file, a process that is also known as raster stacking or layer stacking. The basic idea is to create a new image that has the same x and y dimensions as these individual images and a third dimension z that represents the different bands.

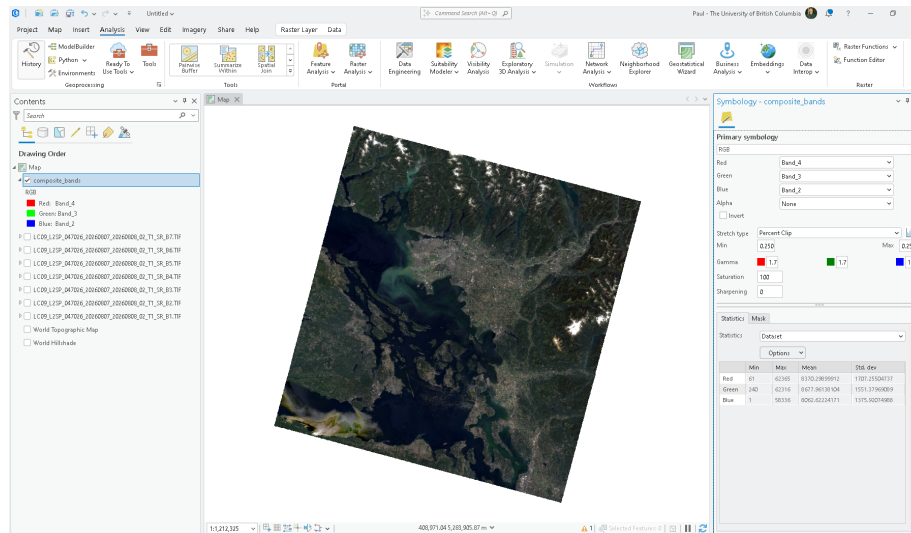
Step 3: Search for the “Composite Bands” tool in the geoprocessing pane. Under “Input Rasters”, click the drop-down menu and select Bands 1-7, in that order, from top to bottom. Name the output “composite_bands” and let it save to your default geodatabase for the project. Do not add any file extension to the file name. Run the tool and inspect the output.



Notice that the new file that is added to your map has some colour, but it is dull.

Step 4: Right-click “composite_bands” in the Contents pane and select “Symbology”. Notice the default symbology type is “RGB”, which refers to the Red, Green, and Blue colour channels of your computer screen. We can map any combination of our 1-7 bands into these three colour channels to produce new colour visualizations. Create a true-colour image by selecting “Band_4” for the drop-down menu next to “Red”, “Band_3” for “Green”, and “Band_2” for “Blue”. This has the effect of mapping the visible red band to the red colour channel, the visible green band to the green colour channel, and the visible blue band to

the blue colour channel. The map automatically updates when you change the band combinations, and this combination is known as a **true colour** image.



Remember that the colours that we see with our eyes are also a wavelength in the electromagnetic spectrum:

- Red – 660 nm (Band 4)
- Green – 560 nm (Band 3)
- Blue – 480 nm (Band 2)

Note that these band number combinations (4-3-2) for true colour imagery are particular to Landsat 9 Operational Land Imager. Other satellite missions may use different band number designations, so when creating true colour imagery you must always pay attention to the bands designated to visible red, green and blue. The table below shows true colour band number combinations for other Landsat missions.

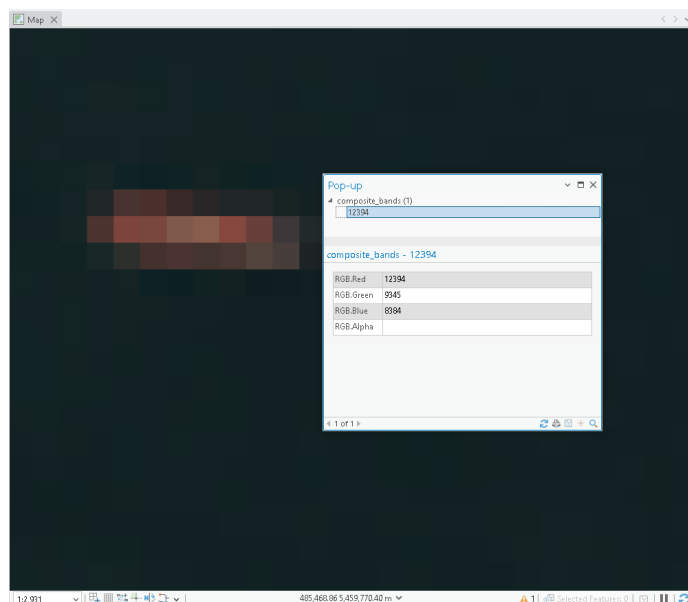
Sensor	Missions	Red	Green	Blue	Band Combination (R-G-B)	
MSS	Landsat 1-5	Band 3	Band 2	-	3-2-1	No true bl
TM	Landsat 4-5	Band 3	Band 2	Band 1	3-2-1	
ETM+	Landsat 7	Band 3	Band 2	Band 1	3-2-1	
OLI / OLI-2	Landsat 8-9	Band 4	Band 3	Band 2	4-3-2	E

In a true colour image, the computer screen visualizes objects the way we see them with our own eyes. In other words, in a true colour image, Landsat Band 2 (Blue – 480 nm) is displayed as blue, Band 3 (Green – 560 nm) is displayed as green, and band 3 (Red – 660 nm) is displayed as red. The mixing of these three primary colours in different intensities (i.e., reflection) produces all of the other colours that we can see.

Recall that each of these band images is just a grey scale image where the brightness of a pixel corresponds to the amount of reflectance that the surface returns to the sensor in that band of the spectrum. The “brightness” of any pixel is recorded using a **digital number**, higher digital numbers are brighter and digital numbers exactly equal to zero are black (i.e., no reflectance). If you look at the grey scale images in the Contents pane of your map (scroll up to the first screenshot in this Task), you will see that each band image has a different range of digital numbers. When we composited these bands together into an RGB image, we still have the same data (digital numbers), but now for every pixel we have three digital numbers (RGB) to display instead of one (single band).

Step 5: Find a feature or land cover in the true colour image with a distinct colour and zoom in closely until the pixels are easy to distinguish, usually less than 1:5000. With your cursor, click on one of the pixels and the resulting pop-up will display the digital numbers that appear in each of the RGB colour channels.

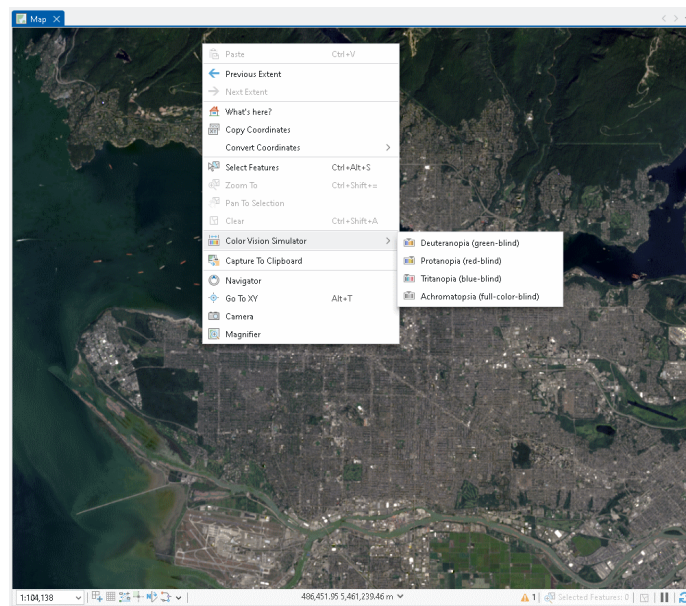
The screenshot below shows one of the many cargo ships that was anchored in English Bay at the time of the image. Notice how the digital number for the red colour channel (Band 4: 12,394) is much higher than the green (Band 3: 8,384) or blue (Band 2: 9,385) colour channels. The result is that the red light-emitting diodes that represents the pixel on your computer screen is brighter than the adjacent green and blue diodes, so the effect is that the cargo ship pixel appears distinctly red in the image.



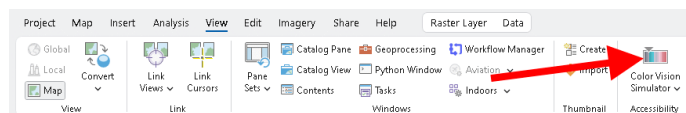
Note that colour vision deficiency affects about 300,000,000 people globally, and

people assigned male at birth are up to 16 times more likely to experience some form of colour vision deficiency compared to people assigned female at birth.

Step 6: You can simulate different colour vision deficiencies in ArcGIS Pro by right-clicking on the RGB image in the map canvas, select “Color Vision Simulator”, and then select any of the types of colour vision deficiencies. This is an especially important consideration when producing accessible remote sensing images and maps for reports or technical documents. If you experience a colour vision deficiency, then you should inform the instructor so that you can receive accommodations for some of the deliverables in this lab.



Step 7: Reset the Color Vision Simulator by navigating to the “View” tab on the top ribbon, and click on the coloured rectangle above “Color Vision Simulator”.



Any band combination where the visible red, green and blue bands are not mapped to the red, green, and blue colour channels, respectively, is called a **false colour** image. False colour images are necessary to visualize electromagnetic radiation that we can not perceive with our own eyes.

Consider the following synesthesia analogy. Your eyes can hear three notes at specific frequencies: R, G, and B. Any combination of these three notes with different volumes, produces all the sounds that you can hear. You look at the true colour pixel of the cargo ship above and you hear a “red” chord. How can

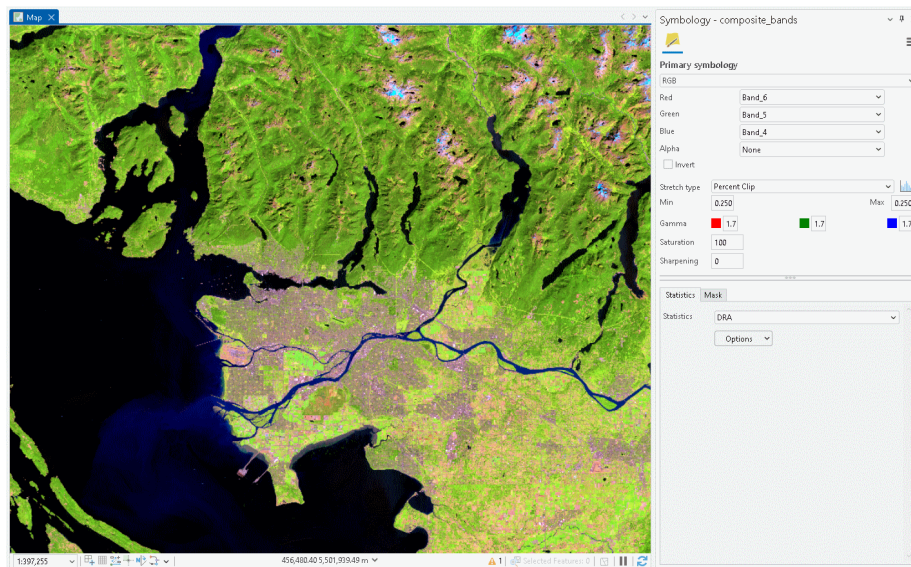
your eyes hear and understand notes that are beyond this range? You build microphones (sensors) that can detect and record these other frequencies (bands) and their volumes (digital numbers) and then you play back the recording in the tones that your eyes can hear with notes of R, G, and B (false colour).

Are you ready to visualize parts of the electromagnetic spectrum that are beyond your natural range of perception?

Step 8: Using what you learned in Step 4, experiment with other band combinations in the RGB symbology of composite_bands and examine all the different false colour images that you can create. Try these R-G-B band combinations in particular:

- 5-4-3
- 4-5-6
- 6-5-4
- 5-6-7

The image below is an example of the 6-5-4 band combination. Note that in the Symbology pane, you can change Statistics from “Dataset” to “DRA Dynamic Range Adjustment”, which will stretch the range and brightness of the colours based on the range of digital numbers currently in view as you zoom in/out and pan around the map. This can make it easier to distinguish different land covers with different colours.



Q8. In the 6-5-4 band combination, match the following cover types to the colors that they appear in the image: healthy vegetation, urban cover, silty water, and clear water. (4 points)

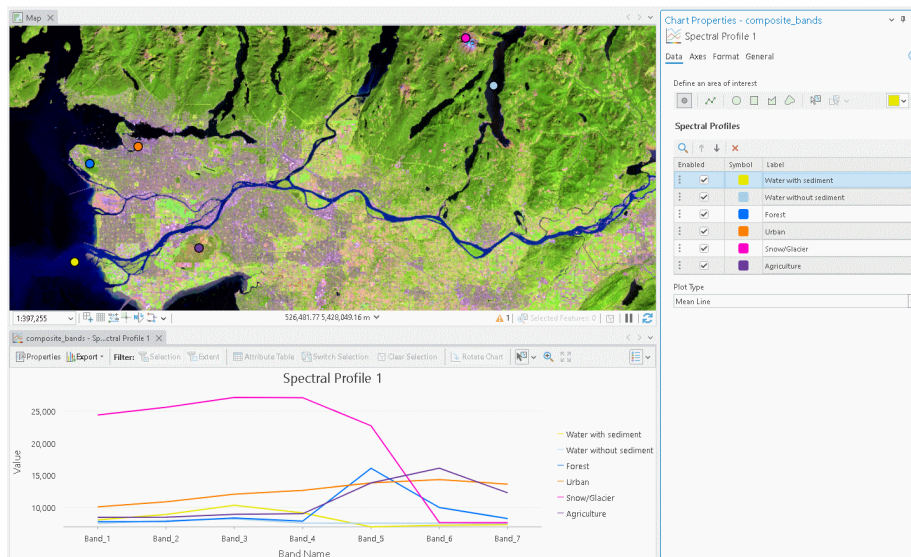
Q9. Which band combination would you use to analyze vegetation? Do some light research on spectral properties and the applications of different Landsat 9 bands and write a sentence or two justifying each of your choices. Then, append a screenshot of your chosen band combination from ArcGIS Pro to your response. (10 points)

Q10. Which band combination would you use to analyze water quality? Do some light research on spectral properties and the applications of different Landsat 9 bands and write a sentence or two justifying each of your choices. Then, append a screenshot of your chosen band combination from ArcGIS Pro to your response. (10 points)

Q11. Which band combination would you use to analyze agriculture? Do some light research on spectral properties and the applications of different Landsat 9 bands and write a sentence or two justifying each of your choices. Then, append a screenshot of your chosen composite from ArcGIS Pro to your response. (10 points)

Q12. Which band combination would you use to analyze urban areas? Do some light research on spectral properties and the applications of different Landsat 9 bands and write a sentence or two justifying each of your choices. Then, append a screenshot of your chosen composite from ArcGIS Pro to your response. (10 points) Next, we will view some spectral signatures across all bands. This is a bit different from visualizing the data with false colour images as our goal now will be to profile the intensity of all seven bands for a given pixel rather than just visualizing colour with three bands. Recall from Task 1 that different land covers can be distinguished by their spectral signatures. You will now use what you have learned from that Task to interpret profiles of pixels from different land covers in this Landsat 9 image.

Step 9: Right-click on the “composite_bands” item in the Contents pane and select “Create Chart”, then “Spectral Profile”. The Chart Properties pane should appear on the right side of your screen and the Spectral Chart on the bottom. In the Chart Properties pane under “Define an Area of Interest” select the point icon and then click a pixel of a land cover that you can see in the image. Repeat this until you have at least six different profiles. Note that it does not matter which band combination that you have displayed, but you might choose a false colour band combination that helps you to find land covers that are visually distinct. You can zoom in closely to place your points precisely. In the chart, you can change the colours of the lines/points to make them easier to distinguish and be sure to add descriptive labels in the Chart Properties pane. You can resize the chart as needed and then export the chart.



Q13. Examine your new spectral profile chart. Compare the different profiles you created to the spectral profiles in Task 1. What is different between them? Why might your spectral curves look different than those above? What does the y-axis of your chart represent? (4 points)

Screenshot 1. Upload a screenshot of your entire ArcGIS Pro window showing your spectral plot. Your ArcGIS account username in the top right must be visible to receive credit. (2 points)

Task 3: Calculating NDVI and Built Up Index

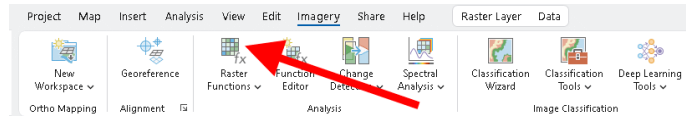
Spectral indices are mathematical equations containing spectral reflectance values from two or more wavelengths used to highlight areas of spectral importance in an image. There are a wide variety of spectral indices used to highlight a variety of different land covers and image properties including burned Areas (Normalized Burn Ratio), urban/ built up areas (Normalized Difference Built-Up Index), and water (Normalized Difference Water Index) to name a few. The Normalized Difference Vegetation Index (NDVI) is a frequently used spectral index that takes advantage of the high near-infrared reflectance and high red absorption properties of healthy vegetation and is therefore often used to quantify vegetation in a remotely sensed multispectral image.

NDVI is calculated with the below formula:

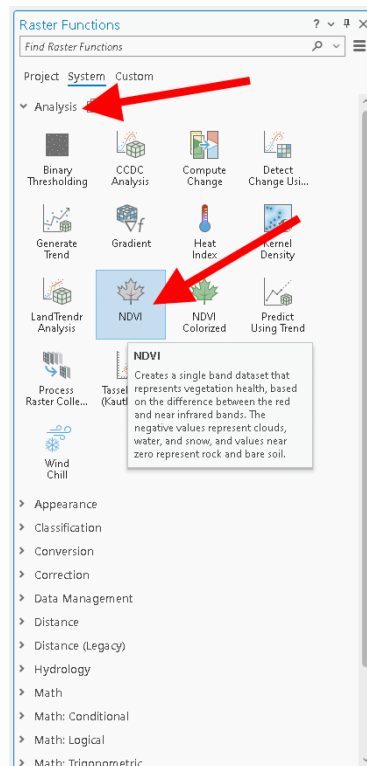
$$NDVI = \frac{(NIR-RED)}{(NIR+RED)}$$

Where NIR is the near-infrared band (Landsat 9 Band 5) and Red is the red band (Landsat 9 Band 4). The results of this equation should be between -1 and 1 with values less than 0 representing water and values between 0-1 representing different levels of healthy vegetation.

Step 1: Navigate to the Imagery tab from the top ribbon and click the “Raster Functions” button.

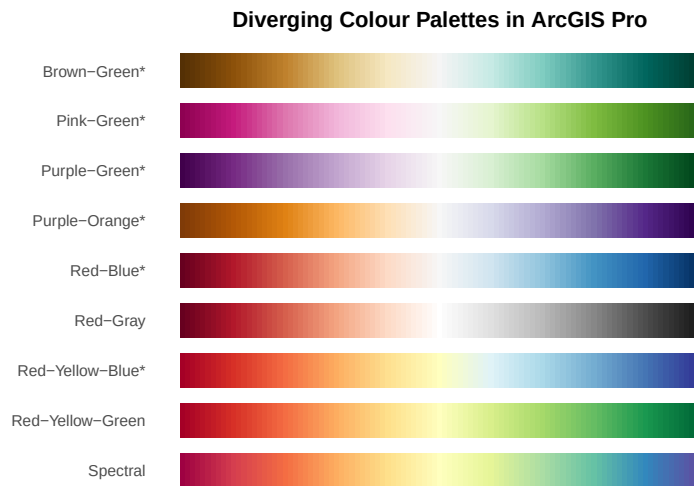


Step 2: The Raster Functions pane should appear. Expand “Analysis” and then select “NDVI” to open.

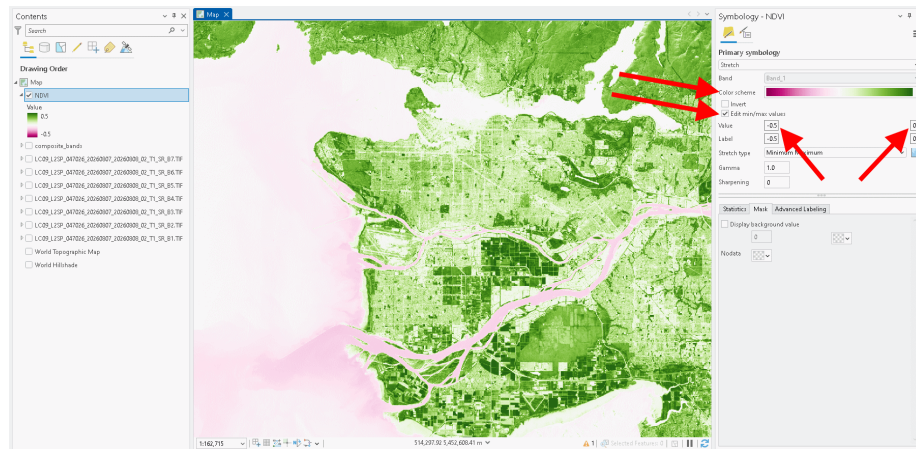


Step 3: The “NDVI Properties” dialogue should appear. Under “Raster” use the drop-down menu and select the “composite_bands” layer. If it is not currently in your map view and can use the folder button and navigate to your lab data folder and select the file from your geodatabase or wherever you saved it in Task 2. “Visible Band ID” corresponds to the red band number and “Near Infrared ID” corresponds to the NIR band number. Select the correct band numbers and then toggle on “Scientific Output” and run the tool.

Step 4: Adjust the symbology of your output NDVI layer so that it uses a diverging colour palette that is stretched between -1 and 1. Right-click the layer in the Contents pane, select “Symbology”, and select a diverging colour from the drop-down list. Most of these will be towards the bottom of the list and will feature either white or yellow as the centre colour. All of the diverging colour palettes are shown in the image below, asterisks indicate palettes that are accessible for people with color vision deficiency.



Step 5: After you have selected a diverging colour palette, toggle off “Invert”, ensure “Stretch Type” is set to “Minimum Maximum”, toggle on “Edit min/max values”, and then in the corresponding boxes just below, select two numbers that are symmetric about zero. In other words, one positive number and one negative number that are equidistant from zero. This symmetry will ensure that the diverging colour palette is applied correctly, so that white or yellow is exactly at the value of zero. The natural range of NDVI is -1 to 1, but this will not always produce the best visualization. The screenshot below shows an example using -0.5 and 0.5, which works well for the Lower Mainland. You may choose different values, based on the local range of NDVI for the area you are visualizing and the types of features that you want to emphasize.



Screenshot 2. Upload a screenshot of your entire ArcGIS Pro window showing your NDVI image with the diverging colour palette applied. Your ArcGIS account username in the top right must be visible to receive credit. (2 points)

Q14. What are the minimum and maximum values of your new NDVI layer? (2 points)

Q15. What do the negative and positive values in the image represent? What about values near or at zero? (4 points)

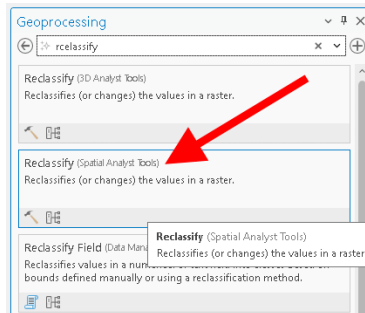
Q16. What information does this type of analysis give us? When and why might this type of analysis be used? (4 points) **Step 6:** Repeat the last step, but this time calculate the Normalized Difference Built-Up Index (NDBI). The NDBI highlights urban areas where there is typically a higher reflectance in the shortwave-infrared region compared to the near-infrared region. The equation is as follows:

$$NDBI = \frac{(SWIR1 - NIR)}{(SWIR1 + NIR)}$$

Task 4: Reclassifying Spectral Indices and Map Making

In this last Task, you will use a technique called “thresholding” and the reclassify tool to extract highly vegetated areas (High NDVI) and urban areas (High NDBI) and then create a map with these layers overlaid with the provided satellite imagery.

Step 1: Navigate to the “Analysis” tab on the top ribbon and select the “Tools” option. The geoprocessing window should appear in the search box type “Reclassify” and select the option with “(Spatial Analyst Tools)” beside it.



Step 2: Under “Input Raster” select the NDVI layer you created in the last Task. The “Reclass field” should automatically become completed and a “Reclassification” table should appear below.

Input raster
NDVI_Layer

Reclass field
VALUE

Reclassification

Start	End	Reverse New Values
-0.276174	-0.062032	2
-0.062032	0.233987	3
0	-0.276174	1
0.233987	0.5	4
0.5	0.939394	5
NODATA	NODATA	NODATA

Unique **Classify**

Step 3: Select the Classify button at the bottom of the table and enter “2” for number of classes. For the first row leave “Start” as is and enter 0.49999 under “End”, change “New” to “NODATA”. For the second row change “Start” to “0.5” and leave “End” as 1 and change “New” to “1”. The Reclassify tool takes the old values from a raster and creates a new raster layer with updated values based on the Reclassification table. In this case we as saying take all pixels with values less than 0.5 and make them NODATA or blank and take all pixels with values greater than 0.5 and make their value 1. Under “Output Raster” save the file as “High_Vegetation_Areas” and press “Run” at the bottom of the window.

Step 4: Repeat the last step for NDBI where all you keep only pixels with values above 0 and change there value to 1. Save this layer as “Urban Areas”.

Map 1. Upload a map that shows your High_Vegetation_Areas and Urban_area layers overtop of the provided Landsat Imagery. Pick appropriate colours and transparency for each layer. (25 points)

Summary

In this lab, you learned how to interpret the electromagnetic spectrum for urban and environmental applications, connect spectral behavior to what you see in imagery, and operationalize those ideas in ArcGIS Pro. You explored Landsat 9 bands, toggled single-band greyscale views, and built both true- and false-colour composites to emphasize specific surface properties. You examined spectral signatures for different land covers and compared them to reference curves to understand why materials separate (or do not) across wavelengths. You then calculated NDVI to highlight vegetation and NDBI to highlight built-up areas, and reclassified these indices to extract high-confidence classes. Finally, you synthesized your results into a clear map that overlays vegetation and urban masks on the base imagery. By the end, you could differentiate EMS regions, justify band choices for specific analyses, quantify patterns with indices, and communicate findings cartographically.

Return to the **Deliverables** section to check off everything you need to submit for credit in the course management system.

Chapter 5

LiDAR for Forest Management

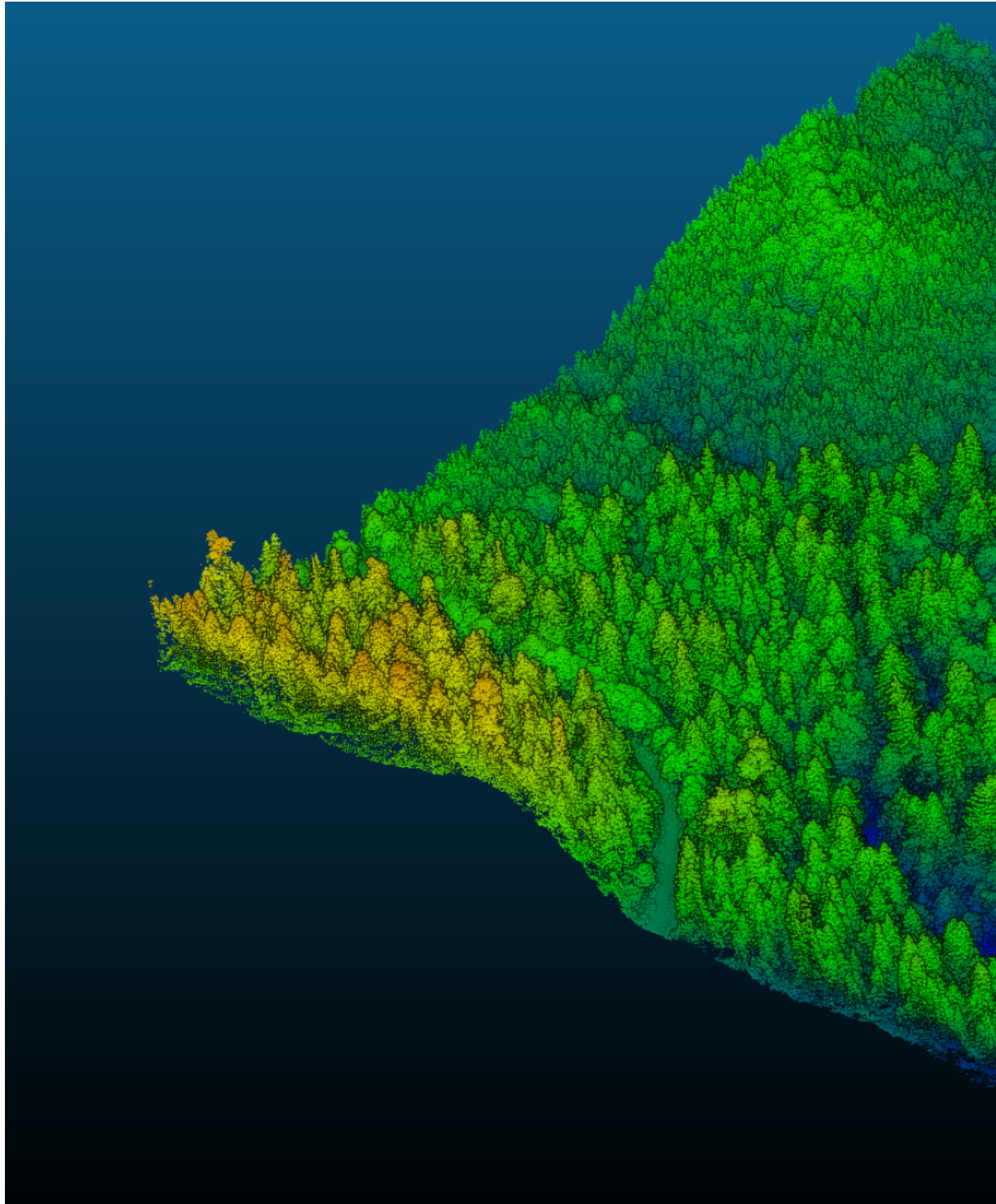
Written by Hana Travers-Smith

Lab Overview

LiDAR (**L**ight **D**etection and **R**anging) is a remote sensing technology that uses laser pulses to measure distances and create detailed 3D maps of objects and environments. To map vegetation, LiDAR emits laser beams from an aircraft or ground-based system towards the vegetation canopy. The laser pulses bounce back upon hitting objects, including leaves, branches, and the ground. By measuring the time it takes for the pulses to return, LiDAR calculates the distance to each point, generating a “point cloud” of data.

“Point clouds” can be used to create high-resolution maps depicting the vertical structure of vegetation, including tree heights, density, and ground cover. These maps are invaluable for understanding ecosystem health, biodiversity, carbon storage, and assisting in land management decisions such as forest monitoring, conservation planning, and urban development.

In this lab you will use high resolution LiDAR data collected by the City of Vancouver to create a model of terrain and vegetation for a forest located on UBC campus.



Learning Objectives

- Understand how LiDAR data is collected
 - Learn how to model terrain and vegetation from a LiDAR point cloud
 - Visualize 3D data in a Scene
-

Deliverables

Answers to the questions posed throughout the lab. (60 points)

Screenshot of your Canopy Height Model (CHM). (15 points)

Screenshot of final 3D ArcGIS Pro Scene. (15 points)

Data

Lidar data collected by the City of Vancouver in .las format

Task 1: Load and understand LiDAR data in a Map Project

Step 1: Download the LiDAR data from the following link: <https://opendata.vancouver.ca/explore/dataset/lidar-2022/information/>

The data is split up into a tiled grid system. You will download one tile covering a section of UBC. Use the “Table” view and search for the following tile and download it: “484000_5455000” by clicking on the .zip link. The zip file is about 2 GB compressed and about 4 GB extracted, so you will need 6 GB of free space on your computer to complete this step. After you have extracted the las file, you can delete the zip file to free up space on your computer.

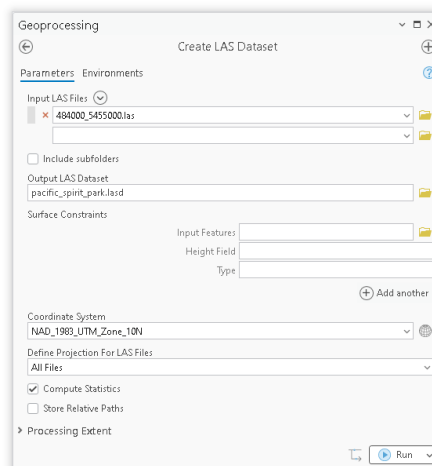
Use the metadata from the City of Vancouver data portal to answer the following questions:

Q1. What is the horizontal and vertical datum of the LAS dataset? Why is it important for Lidar data to have both horizontal AND vertical datums? (10 points)

Q2. What is the point density of the dataset (points per m²)? (4 points) **Step 2:** Create a new project in ArcGIS Pro.

ArcGIS Pro has several tools that we can use to view and analyse LiDAR point clouds. In order to view the dataset, we need to import it as a LAS Dataset. From the geoprocessing pane, search for the “Create LAS dataset” tool, then parameterize it as follows:

- Input File: 484000_5455000.las
- Output LAS Dataset: pacific_spirit_park.lasd
- Coordinate System: Use the horizontal datum from the dataset specifications
- Create PRJ for LAS Files: All LAS files
- Check the “Compute Statistics” box.
- Surface Constraints can be left blank



Step 3: Depending on the zoom extent, you may only see the red bounding box of the las file; this is not an error, you just need to zoom in to see the actual points.

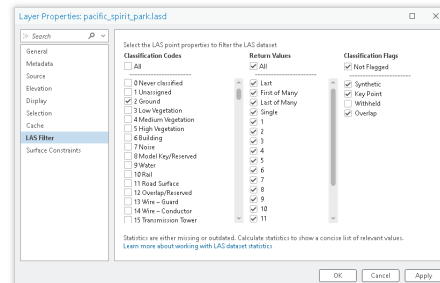
The points can be classified into the following categories:

- Ground
- Non-Ground
- 1st Return (highest feature)

The default display is that no point cloud filters are applied. To quickly filter only ground points right-click on the file in the Contents Pane, navigate to “LAS Filters”, and click “Ground”.

There is also a more detailed classification system, which includes vegetation, water, noise, buildings etc. Right-click the file, select “Properties”, then select “LAS Filter”. This menu gives you more control for which points you want to

display from the dataset.



Task 2: Create a DEM, DSM and CHM

In this task we will create 3 raster datasets by interpolating heights from the point cloud to create continuous surfaces:

- Digital Elevation Model (DEM) - sometimes referred to as a Digital Terrain Model (DTM)
- Digital Surface Model (DSM)
- Canopy Height Model (CHM)

Q3. What does a DEM, DSM and CHM represent? How do you interpret the values in each one? (15 points) **Step 1:** First, we will create a DEM from the .lasd point cloud. **Filter the point cloud so that only points labelled “02 Ground” are displayed.**

Analysis , Tools , and Search for “LAS Dataset to Raster”. We will use the “Binning” method to interpolate elevation. This method works by dividing the point cloud into cells (pixels) and assigning a value to each cell based on the heights of all the points in the cell. For example, cell values could be assigned as the average height of all points within a 10x10m pixel.

- Input LAS Dataset: pacific_spirit_park.lasd
- Output Raster: DEM
- Value Field: Elevation
 - Interpolation Type: Binning
 - Cell Assignment: Average
 - Void Fill Method: Linear
- Output Data Type: Floating Point
- Sampling Type: Cell size
 - Sampling Value: 10 (this defines the resolution of the output raster as 10m)
 - Z factor: 1

Q4. What is the minimum value of the DEM? (2 points)

Q5. What is the maximum value of the DEM? (2 points) You should now have something that looks like this:



Step 2: Next, we will create a DSM. We need to ensure that all points labelled as “Noise” are filtered out from the dataset, we will also filter out “Ground” points and just retain points coming from vegetation/infrastructure. Navigate to the LAS Filter menu in the data Properties tab.

Uncheck the following codes:

- 0 Never Classified
- 1 Unassigned
- 2 Ground
- 7 Noise

Once you have correctly filtered the point cloud open the “LAS Dataset to Raster” tool. This time we will use the “Maximum” value as the Cell Assignment. Name the output raster “DSM”. All other settings can stay the same as the DEM. Click “Run”.

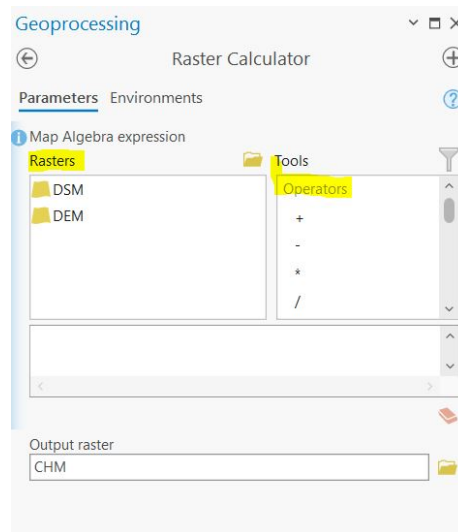
Q6. Why might we want to use the Maximum value as the Cell Assignment method for a DSM? (5 points)

Q7. What features or land cover types can you identify from the DSM? (5 points)

Step 3: Finally, we will create a CHM using the DEM and DSM. Navigate to “Analysis”, “Tools”, and “Raster Calculator” (Spatial Analyst Tools). This tool allows you to create a new raster by combining multiple rasters using simple mathematical operators (adding, subtracting etc). You double-click the layer names from the list to add them to the calculator window.

Calculate the CHM as: $\text{DSM} - \text{DEM}$

Q8. Explain why we need to subtract the DEM from the DSM to calculate canopy height. Describe potential sources of error in deriving a CHM from this method. (10 points)



Step 4: Once you have the CHM raster in ArcGIS, right-click the layer and open Symbolology. From the top, change “Stretch” to “Classify”, then select 3 classes, and change “Method” to “Manual Interval”. Now on the bottom of the Symbolology panel, you can adjust the upper values of each class break to separate unvegetated (< 2 m), medium height vegetation (< 30 m), and high vegetation (> 30 m).

Screenshot 1: Upload a screenshot of your final CHM. Change the default symbology to show unvegetated areas < 2 m, medium vegetation < 30 m, and tall vegetation greater than 30 m. (15 points)

Task 3: Mapping Tree Tops

Using what you learned in the last task, now you will create a point shapefile of treetops using a higher resolution canopy height model.

Step 1: First, we will derive a DSM using points representing High Vegetation. Open the LAS Filter menu and make sure only points labelled “5 High Vegetation” are checked.

Next, use the “LAS Dataset to Raster” tool to create a raster from the filtered point cloud with the following properties:

- 1m spatial resolution

- the raster values should represent the highest point in each cell
- name the output “DSM_1m”

Step 2: Next, produce a DEM “DEM_1m” at the same spatial resolution of 1m using what you learned in the last task.

Step 3: Again following what you learned in the last task, subtract the 1m DEM from the 1m DSM and name the output file **TreeTops**.

Q9. What features are now visible in the 1m DSM that were not visible in the 10m DSM? (5 points) **Step 4:** Next we will use the “Focal Statistics” tool to identify the maximum height of tree crowns across the forest. We will use a Circular Neighborhood with a Radius of 5. This will calculate the maximum elevation observed within a 11m circular moving window (5m is the radius, 10m is the diameter, plus one more cell for the focal cell = 11m). Use the following parameters:

- Input raster: TreeTops
- Output raster: TreeTop_max
- Neighbourhood: Circle
 - Radius: 5
 - Unit type: Cell
- Statistic Type: Maximum
- Ignore no data in calculations: Checked

Q10. What is a moving window? If you are using a focal maximum, explain how cell values assigned in the final output. (5 points) **Step 5:** Open the “Raster Calculator” tool. We will use this tool to find the pixels in the “TreeTops” raster that match the maximum focal height in the “TreeTops_max”. To do this, we will use a True/False conditional statement using the **Con** syntax:

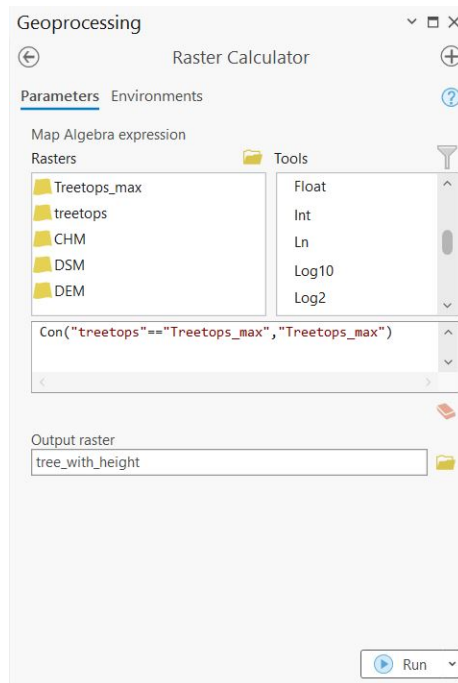
Con(statement, iftrue, if false) - Essentially for each pixel the statement is evaluated and if it is true an action is taken, and if it is false a different action is taken.

Enter the following statement in the “Raster Calculator” tool:

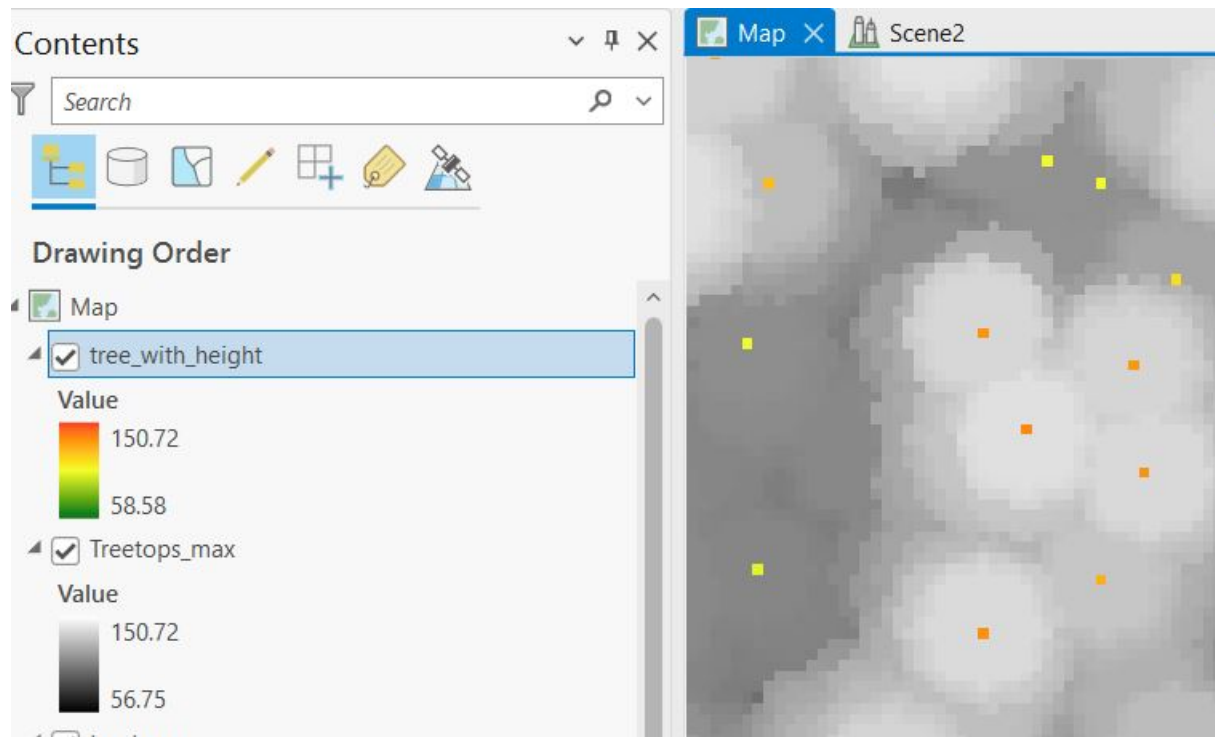
```
Con("TreeTops"=="TreeTops_max", "TreeTops_max")
```

Basically, this is a calculation that evaluates the statement “where is the “TreeTops” raster equal to the maximum elevation value identified from the focal statistics?” Where these pixels match, write the maximum value to the output. Otherwise, write a value of NoData to the output. Save the output as “tree_with_height”.

Refer to the screenshot below for writing this statement correctly.



Inspect the output. Zooming in reveals what we have done. Pixels that represent the maximum height in each focal window are assigned a value equal to the maximum height, while all other pixels have a NoData value.



Step 6: Open the “Raster to Point” tool. The input raster is the raster that you just made. Field is Value, and the output name should be “tree_with_height”.

- Input: tree_with_height
- Field: Value
- Output: Final_treetops

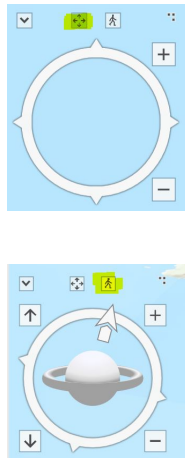
Q11. How many tree tops are in the final output? (2 points)

Task 4: Visualization in 3D

So far we have worked with the point cloud data in flat, 2-dimensional space. In this task, you will explore the point cloud in 3-dimensions in a Scene.

On the top ribbon, select “Insert”, then “New Map”, then “New Local Scene”.

Step 1: Practice using the On Screen Navigator to manipulate the scene. The first toggle lets you navigate in the normal directions and the second toggle (the person) lets you rotate the scene in 3D.



Step 2: Navigate to the UBC campus study area and change the Basemap to “Imagery” in the Map tab located on the top ribbon. In the Contents pane, right-click Ground, located below the Elevation Surfaces group layer. Click “Add Elevation Source”. Browse to the location of your 10m DSM and select it. You can now start to see how the surface features have been incorporated into the surface. Make sure to zoom in to look at areas with high relief and roads.

Add the “TreeTops_Final” point shapefile to the Scene.

Step 3: Click on “Ground” underneath the Elevation Surfaces tab. On the top ribbon click “Elevation Surface Layer”. Set the Vertical Exaggeration to 2 - this will increase the contrast between high/low elevation in the scene. Finally, turn off the WorldElevation3D/Terrain3d layer to better see how the DSM.

Screenshot 2: Upload a screenshot of the final Scene. Add the tree tops points to the visualization. (See example below, note that yours will also include the tree tops points). (15 points)

Q12. Experiment with the 10m and 1m DSM. How does the DSM resolution impact the 3D visualization? Which visualization is more realistic? (5 points)



Summary

In this lab, you learned how LiDAR data is collected, structured, and used to model forested landscapes. You explored raw point cloud data in ArcGIS Pro, applied filters to distinguish between ground, vegetation, and other surface features, and generated continuous surfaces including a Digital Elevation Model (DEM), a Digital Surface Model (DSM), and a Canopy Height Model (CHM). By comparing these layers, you practiced interpreting terrain, vegetation structure, and potential sources of error in canopy height estimation.

You then worked at a finer resolution to map tree tops by applying focal statistics and raster-to-point conversion, creating a dataset of individual tree crown positions and heights. Finally, you visualized the forest in 3D using ArcGIS Pro's Scene tools, experimenting with different resolutions to see how they influence realism and detail. Through these steps, you developed practical skills in working with LiDAR point clouds, learned how to transform raw 3D data into meaningful forest management layers, and gained an understanding of how spatial resolution impacts ecological interpretation.

Return to the **Deliverables** section to check off everything you need to submit for credit in the course management system.